

Sauerhoff Residence

(New Construction)

Finish Specifications

(including updated Design thru 03/18/2023)

as updated: March 20, 2023 5:00 PM

Between the
Developer:

Clifton Avenue Development, LLC
Marblehead, Massachusetts 01945

And the Buyer:

Mitch and Ronna Sauerhoff

For the Project:

7 Ware Lane
Marblehead, MA 01945

Division 00 00 00. Procurement and Contracting Requirements

00 00 00 - Procurement and Contracting Requirements

00 10 00 - Instructions

00 26 00 - Procurement Substitution Procedures

Division 01 00 00. General Requirements

01 50 00 - Temporary Facilities and Controls

01 51 00 - Temporary Utilities

01 51 13 - Temporary Electricity

01 51 23 - Temporary Heating, Cooling, and Ventilating

01 51 26 - Temporary Lighting

01 51 36 - Temporary Water

01 52 00 - Construction Facilities

01 52 19 - Sanitary Facilities

01 53 00 - Temporary Construction

01 54 00 - Construction Aides

01 54 23 - Temporary Scaffolding and Platforms

01 55 00 - Vehicular Access and Parking

01 56 00 - Temporary Barriers and Enclosures

01 57 00 - Temporary Controls

01 58 00 - Project Identification

01 60 00 - Product Requirements (Scope of Work)

01 64 00 - Owner-Furnished Products

01 70 00 - Execution and Closeout Requirements

01 71 00 - Local Conditions

01 74 00 - Cleaning and Waste Management

01 76 00 - Protecting Installed Construction

Division 03 00 00. Concrete

03 00 00 - Concrete

03 10 00 - Concrete Forming and Accessories

03 15 13 - Concrete Accessories

03 21 00 - Reinforcing Steel

03 22 00 - Welded Wire Fabric Reinforcing

03 30 00 - Footings

03 30 01 - Slab Foundations

03 30 02 - Poured Concrete Basement Walls

03 35 00 - Concrete Finishing

03 50 00 - Cast Decks and Underlayment

03 60 00 - Grouting

Division 04 00 00. Masonry

04 00 00 - Masonry

04 05 13 - Masonry Mortaring

04 05 19 - Masonry Anchorage and Reinforcing

04 05 23 - Masonry Accessories

04 40 00 - Stone Assemblies

04 43 00 - Stone Masonry

04 57 00 - Fireplaces

Division 05 00 00. Metals

05 05 23 - Metal Fastenings

05 10 10 - Structural Metal Framing

05 50 00 - Metal Fabrications

Division 06 00 00. Wood, Plastics, and Composites

06 10 00 – Rough Carpentry

06 11 00 - Wood Framing

06 16 00 - Sheathing

06 16 23 - Subflooring

06 18 13 - Glued-Laminated Beams

06 20 00 - Finish Carpentry

06 22 00 - Millwork

06 40 00 - Architectural Woodwork

06 41 00 - Architectural Wood Casework

06 42 00 - Wood Paneling

06 43 00 - Wood Stairs and Railings

Division 07 00 00. Thermal and Moisture Protection

07 00 00 - Thermal and Moisture Protection

07 10 00 - Dampproofing and Waterproofing

07 13 13 - Felt

07 20 00 - Thermal Protection

07 24 00 - Exterior Insulation and Finish Systems

07 30 00 - Steep Slope Roofing

07 31 13 - Asphalt Shingles

07 31 16 - Metal Flashing

07 46 00 - Siding

07 60 00 - Flashing and Sheet Metal

07 71 23 - Manufactured Gutters and Downspouts

07 72 00 - Roof Accessories

07 92 00 - Joint Sealants

Division 08 00 00. Openings

08 00 00 - Openings

08 11 01 - Exterior Doors

08 11 63 - Metal Screen and Storm Doors and Frames

08 14 00 - Interior Doors

08 36 00 - Garage Doors

08 50 00 - Windows

08 51 66 - Metal Window Screens

08 71 00 - Door Hardware

08 71 01 - Door Hardware - Thresholds

08 75 00 - Window Hardware

08 80 00 - Glazing

08 83 00 - Mirrors

Division 09 00 00. Finishes

09 00 00 - Finishes

09 29 00 - Blueboard & Plaster

09 30 00 - Tiling

09 60 00 – All Flooring

09 64 00 - Wood Flooring

09 68 00 - Carpeting

09 70 00 - Wall Finishes

09 90 00 - Interior Painting

09 91 13 - Exterior Painting

09 91 23 - Interior Painting

09 93 00 - Interior Wood

Division 10 00 00. Specialties

10 28 16 - Bath Accessories

10 28 19 - Tub and Shower Doors

10 28 23 - Laundry Accessories

10 30 00 - Fireplaces

10 31 13 - Manufactured Fireplace Chimneys

10 55 00 - Postal Specialties

10 57 00 - Wardrobe and Closet Specialties

Division 11 00 00. Equipment

11 00 00 - Equipment

11 31 00 - Residential Appliances

Division 12 00 00. Furnishings

12 00 00 - Furnishings

12 20 00 - Window Treatments

12 21 00 - Window Blinds

12 22 00 - Curtain and Drapes

12 23 00 - Interior Shutters

12 24 00 - Window Shades

12 30 00 - Casework

12 36 00 - Countertops

Division 22 00 00. Plumbing

22 00 00 - Plumbing

22 40 00 - Plumbing Fixtures

Division 23 00 00. Heating, Ventilating, and Air Conditioning (HVAC)

23 09 00 - Instrumentation and Control for HVAC

23 31 00 - HVAC lines

23 54 16 - Fuel-Fired Furnace

23 70 00 - HVAC Equipment

23 83 00 - Radiant Heating Units

Division 26 00 00. Electrical

26 00 00 - Electrical
26 05 00 - Common Work Results for Electrical
26 05 19 - Conductors and Cables
26 24 00 - Switchboards and Panelboards
26 27 26 - Wiring Devices
26 50 00 - Lighting

Division 28 00 00. Electronic Safety and Security

28 30 00 - Detection and Alarm
28 31 46 - Smoke Detection Sensors

Division 31 00 00. Earthwork

31 10 00 - Site Clearing
31 11 00 - Clearing and Grubbing
31 20 00 - Earth Moving
31 22 00 - Grading
31 22 13 - Rough Grading
31 22 19 - Finish Grading
31 23 00 - Excavation and Fill
31 23 16 - Excavation
31 23 23.13 - Backfill
31 25 00 - Erosion and Sedimentation Controls
31 31 16 - Termite Control
31 40 00 - Shoring and Underpinning
31 50 00 - Excavation Support and Protection

Division 32 00 00. Exterior Improvements

32 01 90 - Operation and Maintenance of Planting
32 01 90.13 - Fertilizing
32 12 16 - Asphalt Driveway
32 13 13 - Concrete Pads and Walks
32 14 00 - Paver Terrace, Front Walk, Bluestone Steps
32 84 13 – Drip Irrigation
32 84 23 - Underground Sprinklers
32 91 13 - Soil Preparation
32 91 13.16 - Mulching
32 91 19.13 - Topsoil Placement and Grading
32 92 19 - Seeding
32 92 23 - Sodding
32 93 00 - Plants

Division 33 00 00. Utilities

33 00 00 - Utilities

Division 00 00 00. Procurement and Contracting Requirements

00 00 00 - Procurement and Contracting Requirements

This document is designed to define specific materials and installation processes used to finish construction at 7 Ware Lane in Marblehead, MA. It was created for use by Brenton Gray Development, LLC for construction specifications and offered as a guide and companion to the Purchase and Sale Agreement with all other construction specification documents and material lists to fully describe in detail this construction project. All information included in this document should be construed as comprehensive and complete as it pertains to all other specifications and plans. The details pertaining to build and finish specifications included herewith supersede any and all other documents, pictures, drawings, and renderings that may have been created for the purposes of sales and marketing purposes or other.

All materials and services to be used to construct 7 Ware Lane will be procured and/or retained through the building company Brenton Gray Development, LLC. No subcontractors, suppliers, designers, or any other service providers will be permitted on-site without prior written authorization by Brenton Gray Development, LLC and Clifton Avenue Development, LLC. Proper insurance including but not limited to liability and workman's compensation policies as governed by the State of MA will be in place and remain current for all subcontractors and/or builders for any/all work associated with this project. No subcontractors, visitors, representatives will be permitted to enter the site without prior written permission and signed releases. Hardhats and safety glasses must be worn at all times when on site by all visitors including Buyers and Buyer's agent(s) and must be accompanied by a representative of Brenton Gray Development and/or Clifton Avenue Development.

00 10 00 - Instructions

Throughout this Specification list there will be text highlighted in [blue](#), where you will see specific information pertaining to your project and the Product Selection process. For instance, under Tiling specifications, if you are specifying a particular color or size, you would need to include additional information about the color such as a sample, swatch, or link to specific details that might include manufacturer, however if your project features a built-in shower unit you could choose to erase or ignore the Tiling specifications information. A companion document summarizes all selections available to the Buyer at the time of P&S. Throughout construction Clifton Avenue Development, LLC representative will be available at regularly scheduled intervals to present samples and mock up to assist with the selection process. These design meetings will be scheduled at a mutually agreed upon time and location including on-site once the premises is safe for such meetings.

Review the product selections provided throughout this document and the Summary document carefully and select only information applicable to the Buyer Selections for construction of your project. Every minute spent on planning will save hours/days at the construction site. If any of these products specified are not to your liking there is an opportunity for you to propose alternatives which the Builder / Development will consider on a case by case basis. If a desired change is accepted following commencement of construction than a Change Order will be issued to memorialize the agreed upon terms.

00 26 00 – Procurement and Substitution Procedures

Alternatives to any products and services as specified will be considered by the Builder and Developer on a case by case basis during a Review Period. Builder and/or Developer and his/her representative(s) may bill \$150.00 per hour for review of alternative products and services including but not limited to specifications for price and availability, compatibility, and any other time deemed necessary to render a final

decision.

Developer and Builder may investigate any proposed products to determine if they are equal or superior in all respects to products specified and coordinate installation of possible accepted substitutions into the Work, making such changes as may be required for the Work to be complete in all respects. Developer or Builder will meet with clients as needed and get Change Order requests signed.

Principals of Developer and Builder may charge \$150.00 per hour for any / all time spent meeting to discuss or researching alternative products and specifications either proposed or requested by Buyers and/or Buyer representative. Re-design time by either Developer / Builder or his/her representatives including architects, engineers, and other professional services rendered necessary to complete the process will be billed at their respective rates.

Division 01 00 00. General Requirements

01 50 00 - Temporary Facilities and Controls

This work shall consist of the application of temporary measures throughout the life of the project.

01 51 00 - Temporary Utilities

All connections and extensions required to provide temporary utilities shall be made by the Developer and Builder at the Developer and Builder's expense.

01 51 13 - Temporary Electricity

Developer to provide and install temporary power for construction site. Marblehead Municipal Light Department will connect to existing power service without disrupting local service requirements. Power feeder service characteristics shall be compatible with the service from which it is taken. Size, type and loading shall be per requirements as established by the National Electric Code (NEC).

The Developer shall provide main service disconnect and over-current protection at a convenient location in accordance with the NEC and Town of Marblehead Municipal Light Department. The Developer or Builder shall provide power outlets for construction operations, with branch wiring and distribution boxes located as necessary and shall provide flexible power cords as required. Provide and install distribution equipment, wiring and outlets to provide single phase branch circuits for power and temporary lighting for work and security.

01 51 23 - Temporary Heating, Cooling, and Ventilating

Builder to provide and install temporary heating and ventilation as needed for construction site depending on weather conditions. Builder to maintain system during construction, while exercising measures to conserve energy. Ventilate enclosed areas to assist cure of materials, to dissipate humidity and to prevent accumulation of dust, fumes, vapors or gases. Supplement with temporary fan units as required to maintain clean air for construction operation.

01 51 26 - Temporary Lighting

Builder to provide and install temporary lighting for construction site following framing of structure. Provide and install temporary lighting in all work areas sufficient to maintain a lighting level during working hours not less than the lighting level required by OSHA standards. As permanent lighting facilities are completed, they may be used in lieu of temporary facilities. Provide temporary lighting as required to satisfy safety and security requirements. Maintain a minimum illumination level of 30 foot-candles measured 3 ft. above floor in areas where finish trades are performing work. At exterior areas, provide

1 foot-candle of light after dark for security purposes.

01 51 36 - Temporary Water

Contractor to provide and install temporary water for construction site. Connect to an existing water source for construction operations.

01 52 00 - Construction Facilities

Sheds shall be portable or mobile buildings, or buildings constructed with floors raised above the ground, securely fixed to foundations, with steps and landings at entrance doors. Structurally sound, secure, weather tight enclosures for office and storage spaces shall be maintained during progress of work and removed at completion of work.

01 53 00 - Sanitary Facilities

Existing facilities shall not be used. Builder to provide and maintain in a neat and sanitary condition such accommodations for the use of his employees and subcontractors as will comply with laws and regulations. Temporary toilet facilities may consist of portable toilets. The number shall be based on number of workers, 1 toilet per 15 workers. Toilet facilities shall be kept supplied and clean and in sanitary condition until the completion of the work and then shall be removed from the site. Upon removal the site shall be properly cleaned and graded. Any visitors to the construction site must be announced at least 24 hours prior to gaining access and all safety precautions must be followed including all current COVID19 precautions such as face masks (COVID19, hand washing, and hard hats while on site.

01 53 00 - Temporary Construction

The Builder shall provide and maintain for duration of work all required temporary stairs, ladders, ramps, runways and hoists for use of all trades.

01 54 00 - Construction Aides

The Builder to provide all construction aids needed during construction which shall include but not limited to; elevators, hoists, cranes, excavators, etc.

01 54 23 - Temporary Scaffolding and Platforms

The Builder shall provide and maintain for duration of work all required temporary standing scaffolding. 'Independent tied' scaffolds will normally be provided for painting, pointing or other maintenance work. 'Putlog scaffolds', used for the construction of brick walls, have only one row of standards which are usually erected some 900mm from the face of the wall, with the boards carried on horizontal members known as 'putlogs'. When used in new construction, the flattened ends of the putlogs are built into the bed joints as work proceeds and then withdrawn on completion, the resulting hole being pointed up.

01 55 00 - Vehicular Access and Parking

Construct and maintain temporary roads or driveways accessing public thoroughfares to serve construction area. Arrange parking areas to accommodate construction personnel and other visitors. When site space is not adequate provide additional off-site parking on street.

01 56 00 - Temporary Barriers and Enclosures

The Builder shall provide temporary barriers to prevent unauthorized entry into construction areas and to protect existing facilities and adjacent properties from damage from construction operations and demolition. Install barricades and covered walkways

required by governing authorities for public right of ways. When necessary install chain link fence around job site.

01 57 00 - Temporary Controls

This work shall consist of the application of temporary measures throughout the life of the project to control erosion and siltation. Such measures shall include, but are not limited to, the use of berms, dikes, dams, sediment basins, fiber mats, silt fences, straw bales, washed gravel or crushed stone, mulch, grasses, slope drains, temporary seeding and other methods. Temporary erosion and siltation control measures as described herein, shall be applied to erodible material exposed by any activity associated with the construction and consistent with state and local control standard.

01 58 00 - Project Identification

Within two weeks after the commencement of work, provide project identification sign at the locations of the temporary electric service. Maintain sign throughout the life of the project or until removed for permanent services. On the sign, list points of contact by name and telephone number.

01 60 00 - Product Requirements (Scope of Work)

All materials shall be installed in strict accordance with the manufacturer's written specifications or Material's Institute Standards. Where the manufacturer's recommended details are used, the manufacturer shall be responsible for the performance of their product. All Items not specifically mentioned that are required to make the work complete and operational shall be included.

Installation and Storage - All materials, supplies and equipment shall be installed per manufacturer's recommendations and per applicable codes and requirements. Material stored on site shall be protected from damage by moisture, wind, sun, abuse or any other harmful effects.

01 64 00 - Owner-Furnished Products

Developer is not responsible for any products furnished by the Buyer. Should Developer or Builder choose to accept any products provided by Buyer that are damaged prior to opening or receiving or that do not fit the Developer shall return them to Buyer immediately. Additional work required to review and or install Buyer furnished products will be charged to the Buyer by the Builder at the appropriate subcontractor's rate. This amount will be billed as a Change Order.

01 70 00 - Execution and Closeout Requirements

The execution of all work shall be in strict accordance with these specifications and manufacturer's written specifications or Material's Institute Standards. Where the manufacturer's recommended details are used, the manufacturer shall be responsible for the performance of their product. All work not specifically mentioned that is required to make the work complete and operational shall be included.

Codes - Construction shall comply with all applicable national, state and local building codes. It is the responsibility of the Developer and Builder to ensure compliance with said codes and modify the specifications and installation techniques as needed to comply with such codes.

Measurements - The Builder shall check and verify all dimensions and conditions before proceeding with construction.

Workmanship - Workmanship shall conform to the best and highest standards of quality

in each trade and shall include all items of fabrication, construction and installation. All work shall be completed by skilled tradesmen and mechanics. Installation of all equipment and materials shall be in strict accordance with manufacturers recommendations.

Insurance - Builders Risk Insurance shall be maintained by the Developer and Builder during the course of construction until final acceptance by the owner. All insurance requirements shall be coordinated with the Buyer according to the P&S. All contractors shall provide and be solely responsible for necessary barricades and safety precautions, and strictly adhere to all governing codes on safety, including the OSHA Act.

Square Footage - Interior finished square footage figures represent heated floor area only and do not include other, unfinished areas including basement "Bonus Space", two-story or vaulted spaces, garage, deck, porch/terrace, balconies or any other unfinished areas. These dimensions are generally measured from the outside face of the stud. Updated design and architectural floorplan and elevation drawings dated March 27, 2023.

3,530 sf	Finished Interior (1st, 2nd fl. incl. 160 sf wine cellar).
675 sf	Bonus Space & Mechanical Rm (unfinished).
576 sf	Parking / Garage.
205 sf	Outside Terrace Area

01 71 00 - Local Conditions

Building conditions vary due to local and regional codes, construction techniques, and weather conditions. These Specifications may not reflect all State and local codes. Clifton Avenue Development, LLC accepts no responsibility for the completeness of these specifications as they relate to code requirements and may update them from time to time without notice. Developer and Builder meet or exceed all State and local building codes but are not responsible for any structural failings due to extreme weather conditions that may occur due to flooding, tornado's, hurricanes, and / or earthquakes.

01 74 00 - Cleaning and Waste Management

Construction site to be in a clean and orderly condition throughout the construction process. Clean interior spaces prior to the start of finish painting and the application of other finishes including flooring. At the conclusion of construction, the project shall be professionally cleaned. This includes but not be limited to; cleaning the interior and exterior glass, surfaces exposed to view, remove temporary labels, stains and foreign substances, polish transparent and glossy surfaces, vacuum carpeted and soft surface areas, sweep and mop all tiled surfaces, etc. Replace filters of operating equipment. Clean equipment and fixtures to a sanitary condition. Remove all waste and surplus materials.

01 76 00 - Protecting Installed Construction

Builder to protect all installed construction. If products or materials come with a protective coating, contractor shall maintain protective coating until construction is complete. Builder shall replace any items that become defective or damaged.

Division 03 00 00. Concrete

03 00 00 - Concrete

All concrete work shall be designed on the basis of "Strength Design" in accordance with ACI 318 "Building Code Requirements for reinforced Concrete." Concrete work shall be proportioned in accordance with ACI 301 "Specifications for Structural Concrete" and ACI 211.1 "Recommended Practice for Selecting Proportions for Normal Weight Concrete". Concrete slabs, patios, driveways, walls and foundations shall be constructed of a minimum 3000 to 3500 psi concrete, 28 day test, with a 4" minimum to 6" maximum slump maximum, air-entrained to 5 - 8%. No additional water shall be added to concrete

after slump test is recorded. Maximum aggregate size shall be $\frac{3}{4}$ ". All aggregates shall conform to ASTM C33. Gravel should be well graded and not exceed 1 $\frac{1}{2}$ " in size. Water shall not exceed 5 $\frac{1}{2}$ gallons for each bag, unless sand is very dry. Concrete shall be mixed using an approved batch machine or mobile mixer until uniform in color and providing a 4" minimum to 6" maximum slump.

03 10 00 - Concrete Forming and Accessories

Provide all labor, materials and equipment necessary for the completion of the plain and reinforced concrete called for on the plans.

Construction of Forms - Construct ICF forms, and of the correct shape and dimensions, constructed tightly and of sufficient strength. Brace and tie the forms together. Make joints and seams mortar tight. Install leakage control materials in accordance with manufacturer's installation instructions.

Form Ties - Use form ties of sufficient strength and in sufficient quantities to prevent spreading of the forms. Place ties at least 1 inch away from the finished surface of the concrete. Do not use ties consisting of twisted wire loops. Leave inner rods in concrete when forms are stripped. Space all form ties equidistant and symmetrical and line up both vertically and horizontally.

03 15 13 - Concrete Accessories

Provide $\frac{1}{2}$ " thick by 4" wide bituminous expansion joint material at all surfaces where slabs adjoin raised slab, crawlspace or basement stem-wall CMU or poured foundations.

03 21 00 - Reinforcing Steel

Reinforcing steel (rebar) shall be minimum ASTM A615, grade 40. All reinforcement splices shall be as follows: #5 bars 25" minimum. All rebar (reinforcing steel) shall be located 2" clear from bottom and side of footing and 2" clear from top. Locate vertical rebar (reinforcing steel) according to plans. All reinforcement splices shall be in accordance with ACI 318 for "Strength Design." All reinforcement steel shall be accurately placed, rigidly supported, and firmly tied in place with bar supports and spacers in accordance with ACI 301 and ACI 318.

03 22 00 - Welded Wire Fabric Reinforcing

Welded wire fabric shall conform to ASTM A105 and be located in the center of the depth. Install at slab on grade conditions.

03 30 00 - Footings

Center all footings on walls, piers, or columns above unless otherwise noted. All footings shall rest on undisturbed virgin soil with minimum soil bearing allowable of 2500 psf, tested for 95 percent compaction, or $\frac{3}{4}$ " stone compacted in 12" lifts to 95 percent density if fill is required. Footings at building perimeter shall be a minimum of 48" below frost line and 24" wide, constructed of 3000 psi concrete. Provide 3 - #5 rebar continuous through footers. Provide #5 rebar corner bars at all corners and intersections of footers, beams and walls. Each side should overlap 2'-0", with a 90 degree bend. Footers shall bear on undisturbed soil and kept free from ground water. Underneath load-bearing walls and interior or exterior column footings, thicken slabs within a 1' radius to 12" thick.

03 30 01 - Slab Foundations

Concrete floor slabs shall be constructed of 3000 psi concrete, 4" thick reinforced with 10 to 12 gauge 6" x 6" welded-wire mesh continuous and rebar as per plans. Place slabs over well-compacted granular fill compacted in 12 inch lifts to 95 percent density per AASHTO T-180 Proctor, and a 4 or 6 mil vapor barrier. Construction or control joints shall

be provided in slabs on grade so that the maximum area between joints shall be 400 sq. ft. and the length of that area is not more than twice the width. Provide smooth steel trowel finish for all interior slab areas and garage surfaces. Provide broom finish texture for all exterior slabs. Slope exterior patio or porch slabs away from building at 1/4" of drop in elevation for every 1'-0" in distance. At garage slab, provide positive drainage and taper lip at garage/overhead door.

03 30 02 - Poured Concrete Basement Walls

Poured walls shall be constructed of 3000 psi concrete with #4 or #5 rebar at 12" x 12" or 18" x 18" on center (OC) placed in a vertical grid. Thickness of foundation walls shall be a minimum of 10" thick and 8" thick at frost walls. Patch all voids and depressions exceeding 3/8 inch in any direction.

Provide peel and stick membrane from ICF manufacturer and install membrane according to manufacturer recommendations as waterproofing system around the exterior perimeter and install drain tile around perimeter as needed.

03 35 00 - Concrete Finishing

Repair of surface defects shall begin immediately after removal of form or pouring of slab foundation. Provide smooth steel trowel finish for all interior slab areas and garage surfaces. Provide broom finish texture for all exterior slabs. Slope exterior patio or porch slabs away from building at 1/4" of drop in elevation for every 1'-0" in distance. At garage slab, provide positive drainage and taper lip at garage/overhead door. Patch all voids and depressions exceeding 3/8 inch in any direction.

03 50 00 - Cast Decks and Underlayment

Install cementitious or fiber backer-board under ceramic tile, marble and stone finishes. Use straight edge as guide to score sheet's face with carbide tipped scoring knife and snap upward along the score line. Large cutouts use a circular saw with carbide tipped blade.

Floor Installation - Install over interior wood or concrete sub-floors. Ensure sub-floor is structurally sound. Ensure the sub-floor is not damaged. Replace any loose, warped or damaged boards. Make certain sub-floor is clean and flat. Exterior grade plywood or hardwood sub-floor should be at least 3/4" in order to provide for a structurally solid, movement free foundation. In addition, the space between the joists should not exceed 24" OC. In any case, the maximum allowable concentrated deflection of your sub-floor may not exceed $L/480$ of the span. Stagger joints. Do not align with plywood joints. Never allow all four corners of sheets to meet at one point. Apply a de-coupling membrane set in mortar or modified thinset to sub-floor per manufacturer's recommendations. Fasten backer-board sheets with proper nails or screws every 8" over the entire surface. Keep the fasteners between 3/4" from sheet edges and 2" in from sheet corners. Provide expansion joints where required.

Countertop Installation - Ensure cabinets are level and secure. Use minimum 1/2" exterior grade plywood or blueboard and plaster positioned across the back of wood cabinets. Space between plywood / blueboard supports is not to exceed 24" OC. Sheet ends and edges must be supported by perimeter framing. Fasten backer-board sheets with proper nails or screws every 8" over the entire surface. Keep the fasteners between 3/4" from sheet edges and 2" in from sheet corners. Provide expansion joints where required.

Wall Installation - Ensure framing is structurally sound. Nominal 2" x 6" wood studs must be straight properly aligned and spaced at 24" OC. In tub and shower enclosures, ensure that the framing is adequately reinforced at the corners. Sheets may be installed vertically or horizontally. Score and snap sheets to required sizes and make necessary cut outs. All joint ends and edges must be supported by a structural framing member or added blocking. In wet areas, install a moisture barrier or waterproof system backer-board. Install sheets 1/4" above floor, tub or shower pan. Fasten backer-board sheets with proper

nails or screws every 8" over the entire surface. Keep the fasteners between 3/4" from sheet edges and 2" in from sheet corners. Set fastener heads flush with the surface, without overdriving. Provide expansion joints where required.

03 60 00 - Grouting

Concrete surfaces to receive grout shall be prepared by removing defective concrete, dirt, oil, grease and other foreign matter to achieve sound, clean concrete surfaces.

Grouting- Mix up grout per manufacturers recommendations. Use a padded grout float to spread the grout over a workable section of tile. Push the grout diagonally across the joints to force it down into the gaps. Grout all the joints except those needed to allow for expansion joints specifically along fixtures, between the floor and walls, or joints in corners in between walls. Expansion joints will be sealed with caulk after the grouting process.

Nonshrink Grout - Lightly roughen concrete surface for maximum bonding per manufacturer's recommendations. Cover concrete areas with protective waterproof covering until ready to place grout. Align and level components to be grouted and maintain in final position until grout placement is complete and accepted.

Division 04 00 00. Masonry

04 05 13 - Masonry Mortaring

Mortar shall be Type "M" or "S" in accordance with ASTM C270, 2500 psi. Grout shall be in accordance with ASTM C270, Type M, 2500 psi concrete using pea gravel for coarse aggregate with a maximum aggregate size of 3/8" and an 8" minimum to 11" maximum slump. Mortar joints shall be 3/8" thick, finished to produce a concave form.

04 05 19 - Masonry Anchorage and Reinforcing

Foundation Ties - Install corrugated metal, or anchor bolts properly sized and spaced as per code. Veneer must be securely attached to the existing construction. Provide one bolt for each 4 ft of wall length. The maximum spacing of bolts or ties, either horizontally or vertically, should not exceed 48 inches for wall plates and 24 inches for wall ties. This tie spacing applies above and below grade. The above grade spacing may be reduced to one tie for each 3 1/4 sq ft of wall area for one and two family dwellings not exceeding one story in height.

Tie Fasteners - Wood Frame - Install corrosion resistant nails to attach the corrugated metal ties to wood frame construction. The nails should penetrate at least 1 1/4" into the wood studs.

Metal - Corrosion resistant, self tapping metal screws should be used to attach metal wire ties to metal construction. Screws should penetrate at least 1/2 inch into the metal structure.

Concrete or Masonry - Install ties with lag bolts and expansion shield or masonry nails. The fasteners and anchors should be corrosion resistant. When installing a continuous steel angle it should conform to ASTM A36 and be treated to resist corrosion. Steel angles for lintels should be a minimum 1/4" thick with at least 3 inch legs.

04 05 23 - Masonry Accessories

Flashing - Flashing materials may be bituminous membranes, plastics, sheet metals or a combination of these. Continuous flashing shall be installed at the bottom of the air space. Flashing must be at or above grade. Flashing should be installed at the heads and sills of all openings and wherever the air space is interrupted. Flashing should extend through the face of the brick veneer to form a drip edge. Where the flashing is not continuous, such as at heads and sills, the ends should be turned up approximately 1

inch.

Weepholes / Weepscreen - Masonry weepholes, each having an area of at least 70 mm (0.1 sq.in.), are required above flashing in masonry cavity wall construction. Weepholes should be located in the head joints immediately above the flashing, and spaced at 24" on center. If open head joints are used, a 24" on center spacing is permitted.

04 40 00 - Stone Assemblies Masonry

Exterior finish chimney, entry post bases, and single riser step at entries shall be random pattern New England fieldstone veneer (Note: front and side entries include single step in to first floor area for easy access). All joints shall be filled with mortar approx 3/8 inch thick unless otherwise noted. Detailing, such as location including entry step riser, chimney veneer, and entry post bases as shown in the construction documents. Stone veneer colors are gray/blue and brown.

04 43 00 - Stone Masonry

Install exterior stone veneer per manufacturer's recommendations including WRB and weep screed for drainage. See construction documents for location and detailing of stone.

04 57 00 - Fireplaces

Install framed opening for gas fireplace and framed chase in Great Room and Master Bedroom. NE fieldstone veneer above roof line over properly framed and supported areas. At gas fireplace area install 3-4" marble on three edges and marble slab at flush hearth all set in thin set. Stone surround veneer shall be placed on either side and directly above fireplaces within mantel/trim area. See construction documents for location of stone above roof line.

Chimney and Flue - Top of Chimney shall be located 3'-0" taller in elevation than any point within 10'-0". Vents shall be U.L. approved and installed as per manufacturer's instructions. If multiple vents are used in the same framed chimney chase, an unequal projection or height an appropriate setback of flue above the stack shall be used to safeguard against exhaust exiting one flue and entering another.

Division 05 00 00. Metals

05 05 23 - Metal Fastenings

Provide 1/2" or 5/8" diameter x 10"-12" long anchor bolts in filled cells and poured concrete walls at 48" on center (OC) maximum at all window locations and on each side of exterior doors. For slabs, install appropriate tie downs or straps as required by applicable building codes.

05 10 10 - Structural Metal Framing

All structural metal for beams and plates shall be in accordance with ASTM A-36. All structural steel for steel columns shall comply with ASTM specification A-53 Grade B or A-501. Structural steel columns shall be 3" minimum inside diameter, unless noted otherwise. All steel details and connections shall be in accordance with the requirements of the latest AISC specifications and latest revisions. Provide all required anchor bolts, bearing plates and metal ties required by standard practice and as noted below.

Tubular Steel shall be in conformance with ASTM A500 Grade B
Steel pipe shall be in conformance with ASTM A-53, Type E or S, Grade A or B.

05 50 00 - Metal Fabrications

Install metal post detailing as specified on construction documents. Install metal gates, grilles, iron work, etc. to meet all applicable building codes, with appropriate detailing and patterns as shown in construction documents. Metal shall be shop built, welded together, cleaned thoroughly and painted with two coats of an anti-rust primer. After installation, apply an additional coat and anti-rust primer in preparation for finish coats.

Division 06 00 00. Wood, Plastics, and Composites

06 00 00 - Wood, Plastics, and Composites

Builder shall review construction documents and provide labor and materials directly or through subcontractors as appropriate pertaining to all work as required in said documents and as specified herein, while complying with all applicable building codes.

06 10 00 - Rough Carpentry

Lumber shall be of live, sound stock and properly dried. Pressure treated lumber shall be used where any lumber shall come into contact with concrete, masonry block or soil and when using as support members for decks, porches or balconies. Lumber for use at exterior shall have a maximum 14 percent moisture content. Provide adequate bracing and shoring during the construction process. Studs and joists cut to install plumbing and/or wiring shall be reinforced by adding metal or wood structural reinforcing to strengthen member back to original capacity and maintain structural integrity. Holes bored shall not be larger than 1/3 the depth and not closer than 2" to the top or bottom of the joist (see manufacturer's guidelines, 2015 IRC, and/or State of MA Ninth Edition).

Wood Species: #2 Southern Yellow Pine, KD SPF, Douglas Fir, etc.

06 11 00 - Wood Framing

Floor Framing - Information below pertains to conventional stick framing, if pre-engineered trusses are used manufactures' guidelines will be followed for installation. Pressure treated lumber shall be used where any lumber shall come into contact with concrete, masonry block, roof curbing or roof blocking.

Girders: Install girders in pockets formed in the foundation or on top of the sill plate. The pocket should allow a minimum of 1/2" on both sides for circulation.

Girders: laminated veneer lumber (LVL), or glue-lam beams.

Sills: Install single 2"x 6" solid pressure treated lumber horizontally on foundation that is in direct contact with concrete. Holes will be bored in sills for anchor bolts. Seal sills for air and insect infiltration.

Floor Joists: Space floor joists 24" on center (OC) or according to load bearing and spanning capabilities of wood species. Joists shall rest on a minimum 2" of bearing wood or 3" of masonry. Cut joists flush with the outside edge of sill. Wood cross bridging shall be at least nominal 1" x 3" lumber with two 8d nails at each end. Install two rows of bridging for 12'-0" and over spans.

Floor Joists: 14" and/or 16" i-joists (TJI or BCI brand) or open web truss at first floor and second floor system as specified by engineers/designers, all at 24" on center (OC).

Exterior Walls - All exterior walls shall be constructed with 2"x 6" wood studs at 24" on center (OC), with single bottom plates and single top plates throughout. Solid blocking at mid-height of all walls over 10'. For exterior corner joints, install at least (3) 2"x 6"s nailed. Where interior partitions meet exterior walls, install 2 studs fastened together with 2"x 6" blocks approximately one foot long. One block is placed at the bottom, one at the top and one about center of the studs.

Where exterior openings occur on 2"x 6" exterior walls, provide structural insulated headers designed with (3) pieces of appropriately sized lumber fastened securely together. At window sills, provide a single piece of 2"x 6" lumber. Provide jack studs as required and based on size openings. Provide 1/2" plywood sheathing at exterior walls with supports at corners for shear wall strength and stiffness.

Fascia and Soffit - Provide and install wood and or PVC fascia and soffit according to plans. Wood or PVC fascia board shall be a 1"x 6" solid pine or PVC set on well blocked sub-fascia structure. For wood or PVC soffit install 3 1/2" crown moulding, 1"x 6" frieze board and louvered or continuous screen soffit vents as required by applicable building codes and roofing manufactures guidelines for ventilation.

Curved / flared roof at main roof area eaves will be installed using rafter extensions of at least 2" x 8" lumber with 3/8" or 1/4" plywood layered to achieve curve around entire perimeter of main roof area. Additional special roof feature to include "eyebrow" feature aka elliptical arched window over kitchen ceiling on west elevation roof. Specific glass window size and design as featured in drawings per Specifications. Eyebrow/ellipse will be framed during roof framing and/or window installation process.

Location: exterior perimeter soffit and mid-level trim band.

Finish, Pattern and Color: semi-gloss finish, continuous molded soffit.

Specify: PVC 1x6 fascia and 1x2 kick out. PVC crown between soffit and freaze.

Color: [White](#), [Off White](#)

Interior Walls

All interior walls shall be wood studs, with single bottom plates and single top plates throughout. Provide solid blocking at mid-height of all walls which exceed 10' in height.

2"x 6" and 2" x 4" stud walls placed 24" on center (OC)

Ceiling & Floor Joists: The size of ceiling joists determined by span, load and grade of lumber. Joists doubled as needed at openings for structural rigidity. Tray ceiling framed over kitchen such that soffits will overhang upper cabinets and allow for crown molding. Upper area of tray will feature cathedral / vault properly positioned to allow eyebrow window light through below. Additional vaulted ceiling installed over Master Bedroom ceiling on 1st floor at rear addition. Ceiling over rear portion of bedroom available to vault only.

2"x 6" or 2" x 8" spaced at 24" on center (OC)

2"x 10" or 2"x12" spaced at 24" on center (OC), attic areas used for storage.

14" or 16" BCI / TJI "I-Joists" as specified during engineering.

Roof Framing: Construction components vary according to geographical location and the size of the overall structure. The size of joists are determined by span, load and the kind and grade of lumber. Joists doubled for structural rigidity at openings as needed. For ridges, hips and valleys, install 2"x 10", 2" x 12", or LVL as needed. Use laminated veneer lumber when ridge, hip or valley spans are greater than 24'-0". Provide fastening anchors at all rafters to plates and at alternate studs.

Roof Decking - Provide and install exterior sheathing of APA rated and code certified 5/8" CDX plywood panels. Sheathing shall be installed with the face grain running across the rafters, vertical joints staggered. Nails shall be 8d galvy ring-shank nails spaced 6" apart on the ends and 8" apart inside. Install with plywood "H" clips between each piece of decking, every 48". Install one layer of moisture barrier 30# felt, overlapped a minimum of 6".

Terraces - Exterior grade lumber shall be used for exterior porch decks and exposed areas. Provide and install galvanized joist hangers to connect 2"x 12" rafters to the main structure every 16" or 24" on center (OC).

06 16 00 - Sheathing

Between sheathing and rigid insulation, install one layer of ASTM approved water resistive barrier (WRB) overlapped according to manufacturer's instructions. Provide and install exterior sheathing of 1/2" rated CDX plywood panels and rigid 1-1/2" thick ISO foam board outboard of WRB. Use 8d galvy ring nails spaced 6" apart on the edges and 12" apart on the studs. For rigid foam board follow EIFS system for specific manufacturers recommendations.

06 16 23 - Subflooring

5/4" tongue and groove extruded aluminum radiant panel plywood sub-floor shall be installed with screws and approved sub-floor adhesive over wood framed joists except below eaves or behind knee walls. Stagger joints.

06 18 13 - Glued-Laminated Beams

For large spans and intermediate beams, structural laminated beams will be used as set forth as needed in the construction documents or by applicable building codes. Laminated timber is hereby defined to include engineered stress-rated products of wood members fabricated from 1" to 2" and up to 6" nominal thickness lumber glued face to face to a depth of four laminations or more.

06 20 00 - Finish Carpentry

All architectural trim and woodwork shall be D grade pine or poplar material suitable for appropriate finishes. Wood that will be stained shall be clear of knots with concealed joints.

06 22 00 - Millwork

Moisture content for interior woodwork shall be 10 percent or less to reduce excess shrinking. Provide and install interior wood trim as per below description. Mock-ups will be provided on-site following close-in during construction for viewing.

Base Molding: 1x8 (3/4"x7-1/4") square edge with WM8455 11/16" x 1-3/8" cap.

Crown Molding: WM8013 11/16" x 3-5/8" and 4-5/8" pine or poplar crown in Master Bedroom (1st & 2nd Floor), Great Room, and Kitchen over upper cabinets, and Dining Room.

Window / Door Casing: Windsor One B200 3-1/2" colonial casing picture frame surround. Windows include 5/4" bullnose window stool and B200 apron. Door include 6/4" x 4" poplar plinth blocks at intersection of door casings and baseboard.

06 40 00 - Architectural Woodwork

Provide and install woodwork as described below:

Fireplace Mantels: (1) site-built, colonial style with pilasters and crown molding below mantel in Great Room.

Paneling: shadow box or raised paneling available in areas such as Stair Hall, Stairway, or Dining Room walls upon request for additional cost.

Built-ins: Mudroom Cabinets/Closets x2 and Mudroom bench. Closet shelf and hanging rod included in all closet finishes. Master Bedroom drawers and built in cabinetry included as per plans. Bookshelves and other built in cabinets including site built or manufactured cabinetry and organization systems available upon request at additional cost.

Wine cellar shelves and wine storage area to carry \$5,000.00 budget allowance. Exact shape and size shelves/storage to be decided during interior design meetings prior to commencement of finish construction in this area.

Glass shelves installed in glass front upper kitchen cabinets to be 3/8" or 1/2" thick tempered glass to ensure adequate support for plates, cups, bowls, etc.

Exterior Fiberglass Columns: 10" round tapered painted columns substitute with stone or shingle pier/pillars at rear terrace.

06 41 00 - Architectural Wood Casework

Provide concealed hinge cabinet hardware all cabinets. Finish to be chrome. Knobs and pull handles to be [round](#), [square](#), or [other shape](#). Cabinet hardware including knobs and pull options (see below) to be provided to Buyer during Interior Finishes Meeting by Builder with all materials procured through Builder or Builder's suppliers.

Specify **Kitchen & Pantry:** Omega Dynasty cabinets inset construction with solid reversed raised panel painted maple doors.

(Door profile options: [Milena](#), [Benson](#), and [Woodward](#))

(Select Color: _____ Door style: _____).

Upper Cab Height: 36" Other _____)

Specify **Powder 1:** Omega Dynasty full access cabinet overlay door. (Door profile options: [Benson](#) or [Woodward](#))

(Select Color: _____ Door style: _____)

Specify **Master Bath 1st Floor:** Omega Dynasty full access cabinet with overlay door. (Door profile options: [Benson](#) or [Woodward](#))

(Select Color: _____ Door style: _____)

Specify **Master Bath 2nd Floor:** Omega Dynasty full access cabinet with overlay door. (Door profile options: [Benson](#) or [Woodward](#))

(Select Color: _____ Door style: _____)

Specify **Bath 2:** Omega Dynasty full access cabinet with overlay door.

(Door profile options: [Benson](#) or [Woodward](#))

(Select Color: _____ Door style: _____)

06 42 00 - Wood Paneling (Exterior)

Provide and install wood or composite paneling in areas as indicated on construction documents, if any. Panel seams shall be butt-jointed without overlapping. Wood / PVC end joints shall match up and be free of defects. All paneling shall be applied over ISO panels on exterior. (Refer to elevation drawings for any paneling detail).

Panel Style: NA

Type: NA

Material: NA

Specify: NA

06 43 00 - Wood Stairs and Railings

Heights of treads, lengths of risers and overall width of stairs shall comply with applicable building codes but shall not be less than 42" wide.

Generally the vertical dimension of one riser and the horizontal dimension of one tread will total 17" or 17.5". Headroom, by most codes, requires a minimum of 6'-8" measured from the front edge of the tread to a line parallel to the stair run. Stair treads shall be constructed of 3/4" thick lumber, risers shall be constructed of 3/4" finish grade lumber, structural stair stringers shall be constructed of 2"x12"s or LVLs. Glue and nail stair assembly on-site or pre-fabricated stairs may be installed. Provide and install railings as shown in construction documents.

Specify Stair Railing: Balustrade component's supplied by Cooper. Flared or curved rail at bottom of stair over newel post or terminate at post TBD.



Stair Treads: match hardwood floors. 3/4" thick treads and 3/4" thick painted risers.

Handrail: Post to post 6008 Somerset sapele mahogany (other species available) ([stain color and precise profile TBD](#))

Balusters: Fitts F5040 Lexington or other stained or painted balusters.

Stair Brackets: skirt board scrolled brackets 1/2" poplar.

Starter Step: install radius tread at first step of stairwell in Stair Hall open right.

Newel Posts: Site built or prefabricate painted or stained recessed panel or radius newel post at base of stairs and square or radius newels every 8' +/-.

(samples of the above components will be provided as requested for above including rail profile and baluster style.)

Division 07 00 00. Thermal and Moisture Protection

07 00 00 - Thermal and Moisture Protection

Developer and Builder shall provide labor and materials pertaining to thermal and moisture protection work, while complying with all applicable building codes.

07 10 00 - Damp proofing and Waterproofing

All joints and penetrations in walls, floors, and roofs shall be made watertight using approved methods and materials. Waterproofing and dampproofing recommendations contained herein generally exceed requirements by code and best practices in building science are applied whenever feasible.

Slab Foundations - Install a polyethylene combined radon / vapor barrier below basement slab, directly underneath concrete. Lap joints according to manufacturer recommendations and seal and tape in accordance with manufacturers guidelines. 4" Schedule 40 PVC combined perimeter drain and radon pipes installed around basement perimeter and vented through roof. Exterior perimeter drain at footing drain into sump basins in basement mechanical area as needed with sealed lids. Sump pumps may added to basins with discharge pipes should the need arise. Lids should remain sealed on top of basins.

Basement Walls - Install ICF Manufacturer's waterproofing membrane to exterior basement walls and foundation surfaces, from a point 12" below the lowest slab to not less than 6" above finish grade. Install as recommended by manufacturers guidelines. Install a minimum 4" slotted drain pipe with a positive outflow around exterior basement wall footings, imbedded in a loose fill gravel, minimum 12" deep. Slotted drain pipe should be wrapped with an appropriate geo-technical fabric to prevent silt buildup. Install other drains necessary for positive site drainage.

07 13 13 - Felt

On all roof surfaces install a minimum #30 asphalt impregnated roofing felt and peel and stick waterproof membrane over eaves and at least 30" over heated areas. Apply aluminum flashing at roof to wall transitions including dormers (at least 3' up roof from outside of exterior wall plane), at eaves, and up rakes as needed and as per code. Fasten all edges with large headed galvanized nails on 6" centers and/or per manufacturer's instructions. Lay courses parallel with eaves.

07 20 00 - Thermal Protection

Effective R values shall be in accordance with or exceed local and state energy codes. Floor, walls and ceiling insulation shall be constructed with either combined batt and blown-in foam or dense pack cellulose. Foam panels 1-1/2" x 4' x 8' poly iso insulation is applied to all exterior vertical surfaces. Install rain screen 1/2" air barrier behind exterior cladding.

All plumbing chases in interior ceilings over living areas shall be wrapped with insulation for sound attenuation.

Exterior Walls - Wall insulation shall be open cell foam inside wall cavity with 1-1/2" ISO rigid foam board on exterior walls with taped seams for air sealing. The R Value of the total combined interior and exterior wall system shall be approximately R-30.

Floors - Between floors, install approx R19 fiberglass batt insulation below radiant heat per manufacturer's recommendations and sound attenuation. R-30 between unfinished/unheated basement and 1st floor area per code.

Roof- blown in closed cell vapor barrier 2"-3" and remaining area filled with open cell with an R value of approximately 49.

Foundation - For concrete slab foundation, provide 2" to 4" thick rigid foam below slab and frost barrier around perimeter of structure. Slope away from building at 1" per foot. Install frost wall footing base 48" below finished grade, along garage/overhead door opening.

07 24 00 - Exterior Insulation and Finish Systems

Apply a 1-1/2" x 4' x 8' polyisocyanurate insulation board to all exterior vertical walls. Attach with approved galvy plastic cap fasteners. Install the base coat of paint to seal red cedar, then finish coat stain shingles. See construction documents for details such as keystones, quoins, horizontal bands, etc.

Trim: 5/4" x 6" window casing with 1-1/4" band bolding and 2" historic PVC sill. PVC rams crown installed at 2nd fl south elevation window heads per plans. Window head flashing installed at all exterior windows and doors white or aluminum finish.

Trim Color paint: [White, Off White, or Other:](#)

Trim 2: 1" x 6" beaded soffit, 1"x5" fascia, 1"x8" freeze board, and 4" PVC crown molding at eaves.

Trim Color paint: [White, Off White, or Other.](#)

07 30 00 - Steep Slope & Low Slope Roofing

Provide and install roof system in accordance with all applicable building codes and manufacturers guidelines. Do not start installation until other trades requiring traffic on roof including framers, carpenters, and masons have completed their work. Do not start installation until vent pipes and other projections through roofs and flashing materials are in place. Main roof slope shall be approximately: 10:12 to 12:12. Lower roof over Master and Sunroom 4:12. Any areas detailed below 4:12 pitch should be completed covered with ice and water shield prior to shingling.

07 31 13 - Asphalt Shingles

Provide and install 30 year composite asphalt architectural shingle over one layer of 30# felt and Grace Ice & Water Shield at eaves, rakes, and valleys. Roof support system to meet dead and live load requirements as specified by manufacturer.

Asphalt Shingle Mfg.: CertainTeed Landmark AR 30 Yr. [Color TBD \(sample board provided at design mtg.\)](#)

07 31 16 - Metal Flashing

Metal step flashing, rakes, and drip edge roof material: white, silver, or other aluminum colors for window and door head flashing, drip edge, rake flashing, and other flashing areas. Copper flashing and gutter upgrade available upon request at additional cost.

07 46 00 - Siding

Provide and install siding exterior in accordance with applicable building codes and manufacturers guidelines.

Wood Siding – Install red cedar R&R 18" tapered shingle siding throughout exterior vertical surfaces per plans. Siding shall be tapered cedar shingle siding with approx. 5-1/2" exposure +/- 3/4". Install a water resistive air barrier (WRAB) between ISO rigid foam and rain screen on all vertical surfaces roof paper and rain screen at exterior walls. When installing siding horizontally, joints should be butted with minimal spacing per manufacturer recommendations.

[18" R&R Red cedar shingles throughout including dormers and gable areas.](#)

Exposure: height: 4-3/4" – 6-1/4"
Siding: machine-sawn tapered shingles.
Detailing: Beveled with approximate 5-1/2" exposure.
Wood species: Red Cedar
Finishes: [Pre-primed. Final solid stain color TBD.](#)
Wood Siding: A or B select.

07 60 00 - Flashing and Sheet Metal

Install appropriate flashing at all joints of chimneys, dormers, walls, vent pipes and other connection points to prevent the infiltration of water. Flashing shall be assembled of 26 gauge minimum galvanized, corrosion resistant sheet metal. Valleys shall be wrapped with Grace Ice & Water Shield and woven shingles. Use 6" x 9" galvanized "step" flashing between wall siding and roof surfaces and between masonry and roof surfaces. Keep flashing concealed except where exposed on vertical surfaces or counter flashing. [Copper flashing at exposed roof and siding details available upon request at additional cost.](#)

07 71 23 - Manufactured Gutters and Downspouts

Install 5" wide white metal aluminum gutters at eaves and 3" round white aluminum or 3" corrugated white downspouts. Attach every 2'-6" (OC) with straps and/or fasteners. Metal should be 25 gauge aluminum.

07 72 00 - Roof Accessories

Vents - Plumbing vents and radon vents installed as needed with proper boot and flashings.

Kitchen Hood Vent Cap - Install galvanized aluminum roof and/or wall vent cap appropriate for size blower and pitch of roof or vertical exterior wall. Locate as specified and as directly over hood blower as possible and as per code.

Wall/Soffit vents - Install perforated aluminum or galvanized metal wall vent caps sized according to manufacturer's recommendations.

07 92 00 - Joint Sealants

Use a 50 year warranty siliconized paintable caulk at high expansion/compression areas, such as around chimneys, tile, ceramic, and around enamel and pre-fabricated tubs and showers. For exterior windows, door frames, interior trim, woodwork and other paintable surfaces use a white Latex based elastomeric caulk over backer rod. Color shall match paint.

Division 08 00 00. Openings

08 00 00 - Openings

Contractor shall review construction documents and provide labor and materials pertaining to the doors and windows as required in said documents and as specified herein, while complying with all applicable building codes. In all sleeping areas provide an operable egress standard window or door directly to exterior as specified in plans.

08 11 01 - Exterior Doors

All exterior doors shall be solid core, insulated and swing inside with weather-tight thresholds and jamb seals. Install weather-stripping around all doors.

Front Door - Fiberglass insulated two panel flat top door with flanking sidelights SDL insulated glass with grilles, and oak threshold. Provide necessary hardware per door schedule below.

Style: Single opening two panel.

Glazing: 5-Lite side panels ea. side w/ insulated clear glazing and grilles.

Material: Fiberglass insulated solid core.

Specify: 3'-6" wide x 6'-8" high door. 1'-6" x 6'-8" ea. Sidelites. (Fiberglass, painted finish or stain finish) (Jeld-Wen)

Great Room - Fiberglass insulated 3/4 View 1-Light Top Down 1-Panel paired opening with sidelights. Insulated glass and sealed oak threshold. Provide necessary hardware per door schedule below.

Style: Paired opening with side lights.

Glazing: 3/4-Light insulated glass clear glazing no grilles doors and sidelights.

Material: Fiberglass insulated.

Specify: 8'-0" wide x 6'-8" high overall opening. (painted finish) (Jeld-Wen)

Sun Room - Fiberglass insulated 3/4 View 1-Light Top Down 1-Panel paired opening. Insulated glass and sealed threshold. Provide necessary hardware per door schedule below.

Style: Paired opening.

Glazing: 3/4-Light panel insulated glass clear glazing no grilles.

Material: Fiberglass insulated solid core.

Specify: 5'-0" wide x 6'-8" high opening. (painted finish) (Jeld-Wen)

Mudroom - Fiberglass insulated 3/4 View 1-Light Top Down 1-Panel paired opening. Insulated glass and sealed threshold. Provide necessary hardware per door schedule below.

Style: Single opening.

Glazing: 3/4-Light panel insulated glass clear glazing no grilles.

Material: Fiberglass insulated solid core.

Specify: 3'-6" x 6'-8" door. (painted finish) (Jeld-Wen)

Kitchen – Fiberglass insulated 3/4 View 1-Light Top Down 1-Panel single door with weather-tight threshold and seals. Provide necessary hardware per door schedule below.

Style: Single door.

Glazing: 3/4-Light panel insulated glass clear glazing no grilles.

Material: Fiberglass insulated.

Specify: 3'-0" wide x 6'-8" painted finish. (Jeld-Wen)

Garage – Fire rated two panel single door with weather-tight threshold and seals. Provide necessary hardware per door schedule below.

Style: Single door.

Material: Fire rated.

Specify: 3'-6" wide x 6'-8" (Jeld-Wen or other)

Combination Screen and Storm doors available upon request at additional cost. Exterior grade aluminum screen/storm combo or mahogany wood frame doors available with [stainless steel](#), [bronze](#), [fiberglass](#) mesh screens. Built-in and removable tempered glass options available. Hardware included by manufacturer for aluminum or match existing doors with wood frames. Exterior door frames prepped for future screens/storms.

08 14 00 - Interior Doors

Interior doors shall be pre-hung with solid-jambs, All interior doors and exterior door trim will be framed with 3-1/2" colonial wood casing on sides and above door frame with 8" x 4" x 5/4" plinth block paint finish at intersection of casing and baseboard. Casing shall be Windsor One B200 3-1/2" prepared for paint finish.

Standard Swing Interior Door - 1-3/4" thick two-panel, solid poplar stiles and rails with solid MDF panels. Door style 210 (see below). Other interior door styles available upon request.

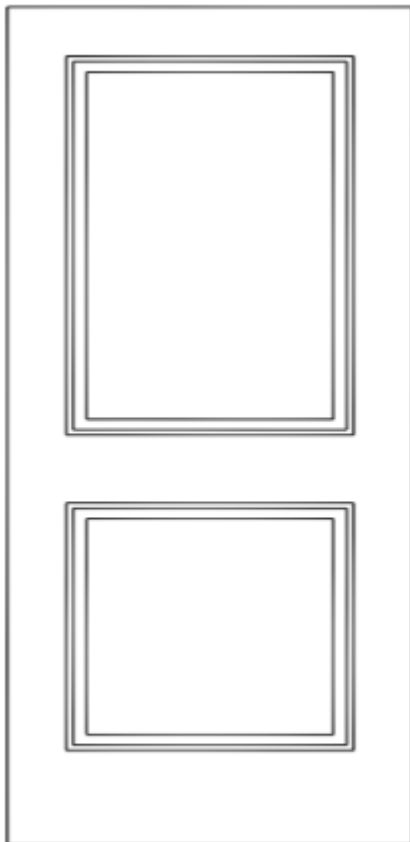
Type: Poplar frame paint grade 1-3/4" thick

Style: Standard two panel (Style 210)

Type: Panel with ogee profile and beaded edge.

Material: solid poplar frame with MDF panel.

Specify: Variety of sizes & swings as per floorplans



210

Pocket Interior Door - 1-3/4" thick two-panel.

Type: Poplar frame paint grade 1-3/4" thick

Style: Standard two panel (Style 210)

Type: Panel with ogee profile beaded edge.

Material: Poplar frame with MDF panel.

Specify: Variety of sizes per floorplans

Bi-fold Interior Door - 1-3/8" thick two-panel, poplar stiles and rails with solid MDF panels. Door style 210. Laundry.

Type: poplar frame paint grade 1-3/8" thick

Style: Standard two panel Style 210

Type: Panel with ogee profile beaded edge.

Material: poplar frame with MDF panel.

08 36 00 - Garage Doors

Garage Carriage Doors - Pair, Garaga insulated steel long panel stamped carriage house style doors. White. Top row 16-Lite glass. R-value 16. All weatherstripping included. LiftMaster 8-1/2 HP electric openers with (2) remotes and photo eyes belt drive including keyless entry pad.

Mfg. & Model: Custom Signature Series

Style: carriage style insulated radius top panel doors.

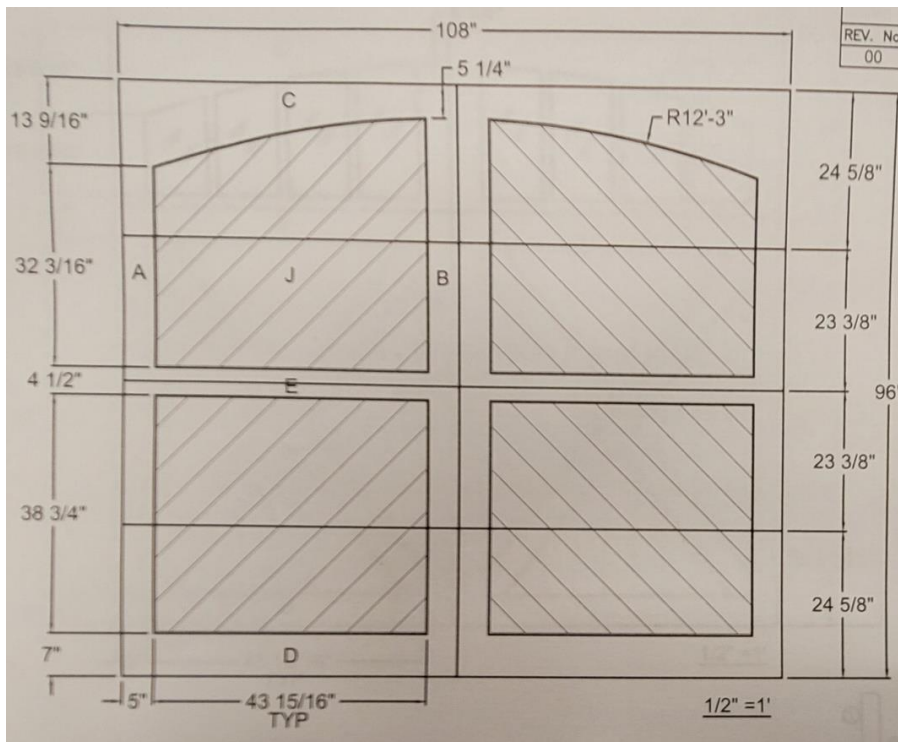
Material: Wood. stained.

Specify: 9'-0" wide x 8'-0" high

Location 1: Garage Entry 1

Location 2: Garage Entry 2

Opener: LiftMaster 8-1/2" HP belt drive w/ 2 remotes and photo eye and keyless entry pad included.



08 50 00 - Windows

All openings to be compliant with applicable building codes concerning egress, lighting, energy performance, and ventilation requirements. Temper all glass located within 2'-0" from exterior doors, within 5' of stairs, and 21" or less off floor, and all glass in doors and above tub enclosures, etc. Provide and install necessary windows and appropriate hardware to operate and lock windows. Bedroom windows shall comply with Code requirements for emergency escape with appropriate egress hardware. Minimum net clear opening shall be 5.7 sq.ft., minimum net clear width shall be 20", minimum net

clear height shall be 24" and sill height shall not exceed 42" above floor. Hardware Finish shall be: [white](#), [bronze](#), [brass](#), [chrome](#). Consult window and glazing schedule below.

Specify windows:

Frame: Harvey Tribute triple glaze (high energy performance).

Style: Double hung, casement, awning, and fixed.

Glazing options: Triple glaze clear glass throughout, Cottage Style (muntins on upper sash only. Upper sash equal size as lower sash.)

Insulation options: Triple Pane Low E

Details: SDL with bar between the glass and grilles, half-screens for DH, full screens for casements.

Low-E coating: Provides additional insulation and shielding.

Gas-Fills: Fill airspace between panes with glass gas to reduce the transfer of heat.

Argon or Krypton filled depending on availability.

Ratings: U value Rating (rate at which heat flows through glass), R-value, air infiltration and shading coefficients all to meet or exceed code requirements.

Interior: interior transom windows to be installed between Master Bedroom 1st fl and Sunroom at approximately 80"-84" off floor for light transmittance.

Window Options:

Lock

Tribute comes with double low profile sash locks, in three finishes, to complement its sleek design. Dual action means you can lock and tilt open the window with a flip of the thumb. Add matching applied lift handle for an upgraded appearance.



COLORS | EXTERIOR



White Vinyl
(Standard)



Almond Vinyl



COLORS | HARDWARE



White
(Standard)



Almond



Oil Rubbed
Bronze



Brushed
Nickel

08 51 66 - Metal Window Screens

Exterior frames shall be a white aluminum screens with smooth joints. Fiberglass mesh. Screens will be installed for easy removal as recommended by manufacturer's guidelines. Any casement window screens will be installed and removable inside.

08 71 00 - Door Hardware

Finish hardware shall include keyed deadbolt locksets at all exterior doors. Interior doors shall be a combination of privacy, passage, and dummy hardware. Hardware shall be as per Schedule. All exterior locksets shall be keyed the same.

Mfg & Model Interior: Emtek. [Georgetown](#) model or [Norwich](#) model or combination thereof with Regular Rosette. Other Emtek models available upon request. All interior doors including pocket and swing doors. All baths and bedrooms receive privacy sets and all other areas receive

passage. Closets receive ball catch and dummy knobs to match all other interior doors.

Mfg & Model Exterior: Emtek. [Willshire](#) keyed entry sets. Mudroom, Kitchen, Great Room, Sunroom, and Front Entry.

Interior Type: Knob or lever door hardware.

Finish: [Brushed nickel](#), [polish brass](#), [chrome](#), [oil rubbed bronze](#), etc.

Door Hardware: Combination of Passage, Privacy, Dummy with Ball Catches.

08 71 01 - Door Hardware - Thresholds

Provide and install anodized aluminum and / or oak thresholds and appropriate door sweeps at exterior doors. Oak threshold may not be available for mudroom fire rated door.

08 72 00 – Door Hardware – Cabinets

*Please note all product photos included in this document are intended to be representative. The products that are eventually installed might be slightly different. Significant differences will be discussed with Buyer prior to ordering and installing.

KITCHEN/PANTRY		
cabinet knobs		
A		1-1/4" "classic" round satin nickel knobs
B		1-3/16" square satin nickel knobs (+\$)
C		1-1/4 round satin nickel knobs (+\$)
drawer pulls		
A		3" satin nickel "cup" drawer pulls

B		3" satin nickel "square" drawer pulls (+\$)
C		3.75" "satin nickel bell" drawer pulls (+\$)
MASTER BATH		
cabinet knobs		
A		1-3/5" round clear bubble glass knobs with satin nickel base
B		1-1/4" round polished chrome cabinet knob
C		1-1/2" round polished chrome cabinet knob
drawer pulls		
A		1-3/5" round clear bubble glass knobs with satin nickel base (used for doors and drawers)

B		3" polished chrome drawer pull
C		2.5" polished chrome "cup" drawer pull
BATH 1		
cabinet knobs		
A		1-9/16" stainless steel round bar knob
B		1-3/8" polished chrome round knob
drawer pulls		
A		3" stainless steel round bar pull

B		3-3/4" polished chrome pull
BATH 2		
cabinet knobs		
A		1-9/16" stainless steel round bar knob
B		1-3/8" polished chrome round knob
drawer pulls		
A		3" stainless steel round bar pull

B		3-3/4" polished chrome pull
1/2 BATHS		
cabinet hardware (choose pull or knob for cabinet doors)		
pull		3" polished chrome cabinet pull
knob		1-1/4" round polished chrome cabinet knob

Finish hardware shall include locksets at all exterior windows. Install as specified by manufacturer.

Type: Sash Lock twist out for casements and sash lock at double hung

Finish: [Brushed nickel, polish brass, chrome, white](#)

Specify: Harvey Tribute

08 80 00 – Glazing (Windows)

Install windows as specified on construction documents per manufacturers recommendations.

Glazing: Clear glass. Tempered (as required, Insulated (low E)

Specify: Harvey Tribute vinyl triple glaze per plans.

Muntin layout: Cottage style (as drawn).

08 83 00 - Mirrors

Builder to supply and install frameless mirrors as per manufacturers recommendations.

Mirror One.

Master Bath

Size: (2) 3'-0" x 5'

Shape: Rectangular

Specify: [Frameless](#)

Mirror Two: Bath 2

Size: 2'-6" x 5'-0"

Shape: Rectangular

Specify: Frameless

Mirror Three: Bath 3

Size of mirror: 2'-6" x 5'-0"

Shape: Rectangular

Specify: Frameless

Mirror Four: Half Bath

Size of mirror: 2'-0" x 4'-0"

Shape: Rectangular

Specify: Frameless

Mirror Six: Half Bath 2

Size of mirror: 2'-0" x 3'-0"

Shape: Oval

Specify: Frameless.

Options: [Kohler K-CB-CLR1620FS substitute or Roburn medicine cabinets upgrade available for additional cost. Mirrors credited toward Roburn medicine cabinet\(s\).](#)

Division 09 00 00. Finishes

09 00 00 - Finishes

Builder shall provide labor and materials pertaining to the finishes as required in said documents and as specified herein, while complying with all applicable building codes.

09 29 00 - Blueboard & Plaster

Interior Walls: Sheath walls and ceilings with 1/2" blueboard and apply plaster finish, either vertically with long edges parallel to framing, or horizontally with long edges at right angles to framing members. Apply one layer of 1/2" x 4' x , 8' or 12' foot lengths to all wall surfaces. Offset joints between layers. Lower level main area and third floor area included. No blueboard or plaster finishes will be applied to basement or garage walls or ceiling as these are considered unfinished areas.

Ceilings: Apply a single layer of 1/2" gypsum wallboard across the supports and fasten with nails or screws. Ceiling finish shall be smooth.

Fire-Rated Gypsum Wallboard: In garage common wall only, around gas water heaters and as required by applicable building codes, install 5/8" Type "X" fire-rated gypsum wallboard.

Water Resistant Gypsum Wallboard or Schluter System Kerdiboard: Around showers, tubs, whirlpools, or as required by applicable building codes, install 1/2" water resistant wallboard.

09 30 00 - Tiling

Tile shall be appropriate grade and finish in accordance with applicable building codes and NTCA guidelines. Builder shall properly clean all surfaces to be covered and install appropriate de-coupling membrane underlayment per manufacturers recommendations on new concrete, wood, and aluminum services. Installation should be carried out with a slow setting cement adhesive, well mixed per manufacturer recommendations.

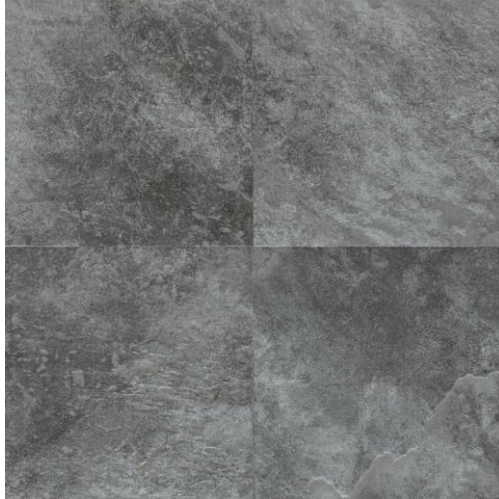
Floors: Contractor shall properly clean all surfaces to be covered and install appropriate underlayment per manufacturers and tile council guidelines.

Mudroom: Floors shall be installed with a 12"x12" gray porcelain, slate, or slate-look tile.

Underlayment: Schluter Systems Ditra

Tile: Daltile Continental or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete. (Color palette available through Builder during design meeting.)



Laundry Room: Floors shall be installed with a 12"x12" porcelain sand matte tile.

Underlayment: Schluter Systems Ditra

Tile: Daltile Cascade or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete. (Color palette available through Builder during design meeting.)



Master Bathroom: Floors shall be installed with a 12"x24" or 12"x12" white marble or porcelain tile.

Underlayment: Schluter Systems Ditra

Tile: Daltile Calacatta Marvel Matte or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete. (Color palette available through Builder during design meeting.)



Master Bath Shower: Floors shall be installed with a 12"x12" gray marble mosaic.

Underlayment: Schluter Systems Ditra

Tile: Daltile Restore Mist or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete. (Color palette available through Builder during design meeting.)

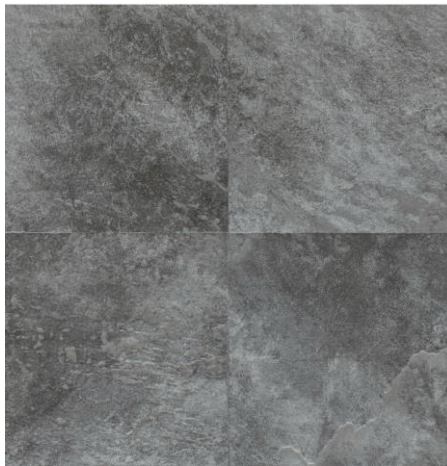


Bath 1: Floors shall be installed with a 12"x12" gray porcelain, slate, or slate-look tile.

Underlayment: Schluter Systems Ditra

Tile: Daltile Restore Mist or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete. (Color palette available through Builder during design meeting.)



Bath 1 Shower: Floors shall be installed with a 12"x12" white porcelain octagon shaped mosaic tile.

Underlayment: Schluter Systems Ditra

Tile: Daltile Hamilton or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete. (Color palette available through Builder during design meeting.)



Bath 2: Floors shall be installed with a 18"x18" marble-look porcelain tile.

Underlayment: Schluter Systems Ditra

Tile: Daltile Lifeproof or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete. (Color palette available through Builder during design meeting.)

Bath 2 Shower: Floors shall be installed with a 12"x12" matte black and white mosaic porcelain tile.

Underlayment: Schluter Systems Ditra

Tile: Daltile or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete. (Color palette available through Builder during design meeting.)

Walls: Contractor shall properly clean all surfaces to be covered and install Schluter System underlayment per manufacturers recommendations.

Master Bath Shower: Walls shall be installed with 12"x12" or 12" x 24" bright white porcelain or marble tile, floor to ceiling height.

Underlayment: Schluter Systems Ditra

Tile: Daltile or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete (Color palette available through Builder during design meeting.)

Bath 1 Shower: Shower walls shall be installed with light gray or other stone or porcelain tile, to a height of 6' above floor.

Tile: Daltile Hamilton or other mfg (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete (Color palette available through Builder during design meeting.)

Bath 2 Shower: Shower walls shall be installed with 3"x9" porcelain subway tile, to the height of 6' high above floor.

Tile: 3"x9" subway (Color, size, and style varieties presented by GC during design meeting prior to material order)

Grout: Laticrete (Color palette available through Builder during design meeting.)

Misc. Areas:

Hearth & Surround Great Room: Install Carrera or like

marble flush hearth, each side and above opening.

Kitchen Backsplash: Included. A variety of materials will be available following initial design meetings to assist in narrowing down options including color, style, size, shape, etc. Installation included and materials will be sourced through GC suppliers. Effort will be made to match Buyer sourced samples as needed.

Sunroom Floors: Included. A variety of materials will be available following initial design meetings to assist in narrowing down options including color, style, size, shape, etc. Installation included and materials will be sourced through GC suppliers. Effort will be made to match Buyer sourced samples as needed.

09 60 00 – All Flooring

Builder shall properly clean all surfaces to be covered and install appropriate underlayment or preparation per manufacturers recommendations. See above for tile and stone floor specifications.

09 64 00 - Wood Flooring

Unfinished: Wood flooring shall be tongue and groove construction, 3/4" thick x 4"-5" wide and 8'-10' lengths with true square edge for a flat smooth surface. Floor adhesive shall be used where stapling around radiant heat tubing is not permitted. Use 2" barbed staple fasteners of approved staples for installing plank or strip flooring. Sand and finish in place, using two coats of oil polyurethane. Hardwood floors can be stained prior to sealing for additional charge.

Area 1st Fl.: Stair Hall & Powder Room, Dining Room, Master Bedroom, Kitchen & Pantry, and Great Room.

Area 2nd Fl.: Bedroom 1, Bedroom 2, and Hall area.

Flooring: Unfinished white oak 3/4" x 4"-5" wide.

Wood species: White oak 4"-5" width & 8'-10' lengths end matched.

Surface color/finish: [Natural clear coat included.](#) [Stain added for additional cost.](#)

Finish: [satin or semi-gloss.](#)

Size: 3-1/4" to 4"-5" white oak.

[* Other hardwood flooring species and finish samples available upon request.](#)

[Additional charges may apply.](#)

09 68 00 - Carpeting

Padding: Cushion padding shall be from Leggett & Platt Arcadia or like cushion not to exceed R-value of 0.80 as recommended by radiant heat panel manufacturer. Pad to be installed in the longest possible lengths. Cushion seams shall not be located directly under carpet seams, and when possible shall be oriented at right angles to carpet seams. All padding shall be smooth, flat and secure with seams tightly butted and taped.

Installation: Carpet shall be installed to produce a tight, smooth, secure and uniform surface. In areas where more than one width of carpet is required, precut the carpet to the proper length allowing for flash up at walls and through doorways. Position first two pieces of carpet, and cut seams by row cutting and trace cutting for the second piece in a manner that will produce a tight, uniform seam requiring a minimum amount of adjusting.

Seams: Carpet seams shall be secured by a non-releasable backed low profile hot melt seaming tape per carpet manufacturers recommendations.

Area: Bedroom 3

Carpet: Home Decorators Collection, [Color TBD](#).

Pad: Leggett & Platt Arcadia or like pad.

Description: PureColor high performance fiber virtually stain-proof coloration and soil-resistant SoilShield. Home Decorators Collection Warranty: 15-year limited stain and soil warranty; 15-year limited wear and tear warranty.

- PureColor Solution-Dyed Fiber provides inherent stain resistance and our exclusive SoilShield technology
- Manufactured from filament fiber to help minimize shedding
- Vacuum and spot clean for maintenance; professional cleaning recommended yearly
- Carpet meets or exceeds CRI Green Label Plus certification standards for interior air quality.

09 70 00 - Wall Finishes

Walls shall be clean and free of defects such as cracks or unfinished joints prior to installation of wall finishes.

09 90 00 - Painting Interior

All nail heads shall be set below the surface and finished smooth. Joints should be taped and covered with a suitable drywall joint compound. Sand the spackled nail heads and joint compound smooth and dust well before priming. Interior walls shall receive a primer coat and two coats of flat or semi-gloss paint. Surfaces shall be sanded before each finish layer is applied. Wallpaper installation available upon request. Wallpaper samples including at least three rolls must be available at time of design meeting to discuss interior finishes.

CEILING

Benjamin Moore Ceiling White (flat)

TRIM

Please select a interior trim color including all doors (Benjamin Moore)

Color:

WALLS

Please select five interior wall colors (Benjamin Moore)

PAINT #1 color:		finish:	
PAINT #2 color:		finish:	
PAINT #3 color:		finish:	
PAINT #4 color:		finish:	
PAINT #5 color:		finish:	

ROOM	PAINT #	WALLPAPER? (+\$)	COMMENTS
Stair Hall / Half Bath			
Master Bedroom			
Dining Room			
Great Room			
Kitchen / Pantry			
Mudroom			
Bedroom 1			
Bath 1			

Bedroom 2			
Bath 2			
Bedroom 3			
Bath 3			
Hallway			
Laundry			

09 91 13 - Exterior Painting

All nail and screw heads shall be set below the surface and finished smooth. Exterior shingle walls shall either receive a semi-solid or solid stain coat (California storm stain) for primed shingles or, for natural shingles, a transparent or semi-transparent stain. All unfinished wood trim pre-primed backside, edges and ends prior to installing. Sand and putty wood surface or peg/epoxy PVC smooth before finish is applied. Surfaces shall be sanded before each finish layer is applied.

Paint/Stain - Prime wood surfaces including faces, edges and ends before installation. After installation, apply at least one coat of wood primer and one coat of finish paint.

FRONT DOOR

Please select exterior door color (Benjamin Moore)

Front door color:

TRIM

Please select a trim color

Ext Trim Color:

SHINGLES

Please select a siding color for a solid color shingle or clapboard finish for the first and second floor

Siding color:

09 93 00 - Interior Wood

Wood surfaces shall be sanded smooth before finish is applied. Fill areas with a wood based filler where nails or other defects appear in the surface.

Paint - Prime wood surfaces including faces, edges and ends before installation. After installation, apply primer and two coats of finish paint. Surfaces shall be sanded before each finish layer is applied.

Balustrade handrails. [Install two coats Minwax or like polyurethane sealer on railing. Railing finished to match hardwood floors.](#)

Specify Railing Stain: TBD.

Division 10 00 00. Specialties

10 28 16 - Bath Accessories

Builder shall install toilet paper rolls and towel bars in all baths. Install toilet paper rolls 15" Above Finished Floor (AFF).

STAIR HALL 1/2 BATH

18" towel bar

Kohler

K-27286-CP

PC

1



single post TP holder

Kohler

K-27292-CP

PC

1



double post TP holder

Kohler

K-37047-CP

PC

1

MASTER BATH

30" towel bar

Kohler

K-11372-CP

PC

1



PC towel ring

Kohler

K-11377-CP

PC

1



PC double post TP holder

Kohler

K-11374-CP

PC

1



12"x12" Schluter kerdi shower niche

Schluter

KB12SN305305A

1

BEDROOM BATH 2



24" towel bar

Kohler

K-37051-CP

PC

1



towel ring

Kohler K-37057-CP PC 1



double post TP holder

Kohler K-37047-CP PC 1

BATH 3



24" towel bar

Kohler K-37051-CP PC 1



towel ring

Kohler K-37057-CP PC 1



double post TP holder

Kohler K-37047-CP PC 1

10 28 19 - Tub and Shower Doors

Shower doors shall be constructed of tempered safety glass with all exposed edges polished and rounded. Swinging doors shall have vinyl seal at both the latch jamb and hinge jamb side of door.

Location: Master Bath: Frameless Glass Door: Clear. Chrome hardware.

Swing out door. Size 26"-30" wide x 72" high.

Location: Bath 2 (aka Master Bath 2nd): Frameless Glass Door: Clear. Chrome hardware.

Location: Bath 3: Kohler clear glass enclosure. Sliding doors.

Frame: Polished chrome frame Door: sliding.

10 28 23 - Laundry Accessories

Builder shall supply and install granite countertop (or other stonelike surface material selected) in 2nd floor Laundry Room directly over washer and dryer. The design drawings locate the appliances side by side unit washer and dryer on left with countertop over them. Hanging rod and shelves over WD and hanging area.

10 30 00 - Fireplace

Pre-fabricated gas fireplaces shall be U.L. approved and installed per manufacturer's specifications. Install units as specified below. Venting for units through chase in fireplace above and chimney through the roof with required offsets.

Manufacturer: Napoleon

Model Number: (2) x Altitude X-36 Series gas.

Unit Dimension: 34" high x 35" wide. Approx.

Location(s): Great Room and Master Bedroom.

Optional (Additional Charge Apply): 3 year extended service, fan, remote included.

10 31 13 - Manufactured Fireplace Chimneys

Chimney for prefabricated fireplace units shall be lined with 8"-10" diameter insulated metal flue as specified by manufacturer. Number of flue sections per manufacturer. Framed chimney with exterior new England fieldstone veneer finish and chimney details as noted in construction documents.

Chimney and Flue - Top of Chimney shall be located 2'-0" to 3'-0" taller in elevation than any point within 10'-0". Flues shall be U.L. approved and installed as per manufacturer's instructions. If multiple flues are used in the same chimney chase, an unequal projection or height of flue above the stack shall be used to safeguard against smoke exiting one flue and entering another.

10 55 00 - Postal Specialties

Builder shall install standard postal mailbox. Metal mailbox to install as per postal requirements with street number and road visible and easy to read. Install mailbox on at front of house near entry or at driveway.

Mailbox: Provided by Buyer and installed by Builder at Street or front entry once confirmed with Postmaster in Town of Marblehead.

10 57 00 - Wardrobe and Closet Specialties

Install wood shelf in all closets as indicated on construction documents. Provide aluminum hanging rod, and supports every 3'-6" – 4'-0" maximum on center (OC) as needed.

Specify: 1st Floor Stair Hall Coat Closet with hanging rod and shelf.

Specify: 1st Floor Mudroom Closet with hanging rod and shelf.

Specify: 1st Floor Mudroom Utility fixed shelving/ cubbies and stand up storage.

Specify: Master Bedrm Closet 1st fl. (2) ea built-in shelf and aluminum hanging rod.

Specify: Bedroom 1 Closet 2nd fl. (2) built-in shelf and aluminum hanging rod

Specify: Bedroom 1 Linen Closet. Built in storage fixed shelves.

Specify: Bedroom 1 Built in drawers. Built in drawer and or shelves.

Specify: Bedroom 1 Built in hanging closet at entry foyer including shelf and hanging rod.

Specify: Bedroom 1 Built in hanging in bedrm area both with shelf and hanging storage area.

Specify: Laundry built-in open shelves over WD and shelf and hanging rod on opposite wall.

Specify: Bedroom 2 Built in hanging rod and shelf above.

Specify: Bedroom 3 Built in hanging rod and shelf above.

*Custom closet designs and built-in storage options available through Builder upon request at additional costs may apply.

Division 11 00 00. Equipment

11 31 00 - Residential Appliances

Install electric and gas appliances as per locations specified on construction documents, including all venting and fresh air supply requirements per manufacturers recommendations and applicable building codes. See electrical specifications for wiring information.

Appliance Schedule:

Qty	Description	Mfg	Model
1	36" Over & Under refrig/freezer w internal dispenser	SubZero	CL3650UID/S
1	36" Dual Fuel Range 6 burners	Wolf	DF36650/S/P
1	24" Dishwasher	Cove	NA
1	30" E Series Pro Steam Oven	Wolf	CSO30PE/S/PH
1	30" M Series Contemp SS Built in Oven.	Wolf	SO3050CM/S
1	42" Pro Wall Hood – 27" depth	Wolf	PW422718
1	Sink disposal	InSinkerator	¾ HP
2	Washer	Maytag	MVW7230HW
2	Dryer	Maytag	MED7230HW

Division 12 00 00. Furnishings

12 21 00 - Window Blinds

Available upon request for additional charge.

Venetian Blinds - Specify manufacturer, color, finish

Horizontal Blinds- Specify manufacturer, color, finish

Vertical Blinds - Specify manufacturer, color, finish

12 22 00 - Curtain and Drapes

Available upon request for additional charge including blocking for future use.

Fabric - Specify manufacturer, color, detailing

Backing - vinyl, fabric, lightproof

Mounting Brackets or Rods - Specify manufacturer, color, detailing

Fastenings - Rings, drapery hooks, (specify finish)

12 23 00 - Interior Shutters

Wood shutters available upon request for additional charge.

Wood / Vinyl - Home Decorators Collection or Hunter Douglas
Color, finish, panels.

12 24 00 - Window Shades

Roman Shades- Specify manufacturer, color, finish

Cellular Shades - Specify manufacturer, color, finish

Roller Shades- Specify manufacturer, color, finish

Pleated Shades- Specify manufacturer, color, finish

12 25 00 - Other

Wire metal open shelving provided for dry storage items in
Pantry area. Shelving exact size and style options will be
available for final selection during interior design meeting.

12 30 00 - Casework

Install pre-fabricated Omega Dynasty in Kitchen, Bathrooms and other built-in cabinets at
specified locations in construction documents. Dimensions of base cabinets shall be: 24"
deep x 34.5" high in Kitchen and 18" to 21" deep in bathrooms. Dimensions of overhead
cabinets shall be: 13" deep x 36" high. [39" high, 42' high, and 48" high upper cabinets
available. Additional costs may apply.](#) Provide concealed softclose hinges and cabinet
hardware as specified in finish schedules.

Pre-fabricated Omega & Custom on-site built-ins: See construction documents.

Finish: Pre-finished by Mtg.

Specify Kitchen: Maple, framed, beaded inset panel.

Door Style: [TBD](#) **Color:** [TBD](#)

Specify Pantry: Maple, framed, beaded inset panel.

Door Style: [TBD](#) **Color:** [TBD](#)

Specify 1st Fl ½ Bath: Maple, frameless, full overlay.

Door Style: [TBD](#) **Color:** [TBD](#)

Specify 1st Fl Master Bath: Maple, frameless, full overlay.

Door Style: [TBD](#) **Color:** [TBD](#)

Specify Bath One: Maple, frameless, full overlay.

Door Style: [TBD](#) **Color:** [TBD](#)

Specify Bath Two: Maple, frameless, full overlay.

Door Style: [TBD](#) **Color:** [TBD](#)

Specify Bedroom 1: Maple, framed, inset panel drawer fronts

Door Style: [TBD](#) **Color:** [TBD](#)

Specify Mudroom: Custom wood built-in bench seat.

12 36 00 - Countertops

Quartzite or Granite counter tops shall be provided and installed in Kitchen / Pantry. Master Bathroom and all other bathrooms including powder room. Counter surfaces shall be fastened to cabinets or plywood as recommended by manufacturer/fabricator and installer. Quartz counter edges shall be installed with eased edge. Additional edge profiles are available through Builder's supplier at additional cost. 4" backsplash provided between counter and wall in Kitchen/Pantry and Bathrooms where tile is not present. Countertop samples available through Builder during Interior Design Meeting process.

Counter Top 1: Kitchen: Granite TBD

Counter Top 2: Pantry: Granite TBD

Counter Top 3: Master Bathroom, Granite TBD.

Edge: Bevel / Pencil / Eased edge.

Counter Top 4: Bathroom 1, Granite TBD.

Edge: Bevel / Pencil / Eased edge.

Counter Top 5: Bathroom 2, Granite TBD.

Edge: Bevel / Pencil / Eased edge.

Counter Top 6: Half Bath, Granite TBD.

Edge: Bevel / Pencil / Eased edge.

Counter Top 7: Laundry, Granite TBD approx. 6' x 30".

Edge: Bevel / Pencil / Eased edge.

Bench Seat 1: Granite TBD match counter

Edge: Bevel / Pencil / Eased edge.

Bench Seat 2: Granite TBD.

Edge: Bevel / Pencil / Eased edge.

* Countertops listed above supplied and installed exclusively through BG supplier. BGD supplier offers extensive selection of different styles and colors of quartz, quartzite, granite, and marble countertops. Substitutions may be possible subject to price and availability.

Division 22 00 00. Plumbing

22 00 00 - Plumbing

Plumbing shall be a fully operational system of hot and cold water. Provide and install all piping, soil, vents, drains, sewage removal and water supply systems to connect with appropriate water and sewage systems for 1st and 2nd floor living areas. Provide and install appropriate insulation around piping surrounding main living areas. All permits and inspections are to be obtained by Developer and Builder and Plumber as required by local building codes and the Uniform Plumbing Code.

Sewer and Waste Piping - Drainage system shall be Schedule 40 PVC pipe. All connections shall have PVC cement or appropriate joint compound and assembled tight for no leakage. Condensate drains for mini-split heads shall be constructed of Schedule 40 PVC or PEX tubing. Valves shall be Milwaukee Brand or equal. Building sewer shall be approved Schedule 40 PVC pipe. Connection to public sewer system shall comply with all local requirements. Caulk joints or provide neoprene gaskets for all sewer lines. Pitch shall be a minimum 1/8" per foot for soil lines larger than 3" diameter and a minimum of 1/4" per foot for soil lines 3" diameter or less. Interior drain/waste piping insulated with batt insulation for sound attenuation.

Water Pipes - From public water line, install Type "L" or "K" 1" copper pipes, located below frost line. From the meter to the building, install 1" supply water lines. Use 1/2" and 3/4" PEX minimum lines from supply lines to each plumbing fixture as required. At water heaters and hose bibs install a minimum 1/2" and 3/4" pipe. From water heater install 1/2" and 3/4" pipe to each room with branches to fixtures as described above. Fittings shall be wrought copper or PEX, soldered with 95-5 solder and suitable flux. Use polished chrome adjustable brass P-traps with wall escutcheons at all exposed locations.

Provide shut-off valves at sinks, toilets, water heater and other fixtures as required. Test all pipes under pressure per building code requirements.

Waste Drainage - Install sewage clean-out at the end of each horizontal drainage run and every 100 feet per building code requirements. Vents shall be installed throughout plumbing connections and connected with the vertical stacks and vented through the roof. Check with local building code officials for specific venting requirements.

Water Heater - Install one 80 gallon Lochinvar or equivalent indirect gas fired hot water heater. Water heater shall be heated by condensing boiler and have appropriate safety valves, back flow preventers, pressure relief valves and drain assemblies. Install Type "X" fire-rated gypsum wallboard surrounding gas water heaters as needed only. Follow manufacturer recommendations and building code requirements for installation and use

Specify: indirect gas fired

Specify Powder: Lochinvar indirect 80-85 gallon storage tank.

22 40 00 - Plumbing Fixtures

Provide necessary piping, water and drains for plumbing fixtures either as shown on the construction documents or listed herein.

Fixtures and Fittings - Provide and install plumbing fixtures as listed below.

Location	Fixtures	Qty	Descrip.	Model
Kitchen	Sink SS	1	32" x18"x9" undermount	Kohler Sterling 847937
Kitchen	Faucet PC	1	Simplice 1 hole deck mount single lever handle with pull down spout.	Kohler 770793
Kitchen	Sink	1	Bar sink. 15" ss	Kohler Vault
Kitchen	Faucet	1	Bar sink. Simplice.	Kohler
Pantry	Sink SS	1	16" x 18" x 8" undercounter prep sink.	Kohler 729057
Pantry	Faucet	1	Simplice lever HDL Prep faucet	Kohler 1851667
Master Bath	Univ Valve	1	Right temp pressure balanced	Kohler 1606501
Master Bath	Master shower	1	2 or 3 way transfer valve.	Kohler 929281
Master Bath	Shwr Fct Trim	1	Shower faucet trim	Kohler 2318710
Master Bath	Shwr transfer valve	1	Trim	Kohler 2325184
Master Bath	Body spray	2	Statement round Adj. 26299-G-CP	Kohler 2313621
Master Bath	Handshow er	1	K-99898 PC Awaken	1886941
Master Bath	Handshow er supply elbow	1	Awaken Supply Elbow K-98350- CP	Kohler 1314254
Master Bath	Kohler 971299	1	White Highline WC comfort height	Kohler 971299
Master Bath	WC seat	1	Cachet plastic seat w cover	K-4636
Master Bath	Sink	1	Undercounter sink	Kohler 20,000-1
Master Bath	Faucet	1	Venza PC sink faucet	Kohler 28125-4-CP

Master Bath	Rain head	2	Rain head x2	Kohler Awaken
Bath 2	Shower	1	Simplice shower	Kohler TS27392-4G-
Bath 2	Sink	1	White oval bowl undermount	Kohler 74490
Bath 2	Bath faucet	1	Simplice PC	Kohler 27390-4-CP
Bath 2	WC	1	Highline 2 piece comfort height elongated WC	K-3999-0 white
Bath 2	Toilet Seat	1	Quiet close elongated	Kohler 4636-RL-0
Bath 2	Shower	1	Rain head	Kohler Awaken
Bath 3	Shower valve	1	Univ rite temp pres bal valve kit	Kohler K-P88304
Bath 3	Shower trim	1	Simplice shower faucet trim	Kohler TS27392-4G-CP
Bath 3	Sink	1	White oval bowl undermount	Kohler 74490
Bath 3	Bath faucet	1	Simplice PC	Kohler 27390-4-CP
Bath 3	WC	1	Highline 2 piece comfort height elongated WC	K-3999-0 white
Bath 3	Toilet Seat	1	Quiet close elongated	Kohler 4636-RL-0
Bath 3	Shower	1	Rain head	Kohler Awaken
Half Bath	Sink	1	White oval Caxton 17x14 undermount	Kohler 2210-0
Half Bath	Faucet	1	Bellera PC 2H WS DM Bath faucet	Kohler 27380-4-CP
Half Bath	WC	1	White Highline 2 piece comfort height elongated	Kohler 3999-0
Half Bath	WC Seat	1	Closed front elongated seat	Kohler 4636-0

Division 23 00 00. Heating, Ventilating, and Air Conditioning (HVAC)

23 09 00 - Instrumentation and Control for HVAC

Install digital radiant thermostats Tekmar 519 on 1st and 2nd floors of house (2). Zone heating by floor and AC by room with mini-split heat pumps. The location of controls shall be coordinated to avoid conflicts with furnishings or uses for the rooms in which controls are located. AC controls mounted at entry of rooms with head units for convenience. Kitchen ventilation is based on size and CFM requirements of hood blowers). If the blowers are rated for 395 cfm or more make up air will be required in the 1st floor cooking area and these requirements have been factored in to ventilation reqi

23 31 00 - HVAC lines, refrigerant lines, and condensate lines

Layouts for lines shall be determined on-site as not to interfere with finished ceilings.

23 54 16 - Gas-Fired Furnace

HVAC unit shall be a high efficiency minimum 85,000 BTU 95% high efficiency gas condensing boiler. The system will contain water distribution pipes, backflow preventers, valves, pumps, and thermostats. Provide necessary venting and fire-rated wall system

around furnace as needed.

23 70 00 - HVAC Equipment - Electric Heat Pump Mitsubishi / Fujitsu / Samsung mini-splits.

AC shall be a fully operational designed to meet individual room needs. All permits and inspections shall be obtained by mechanical contractor as required by applicable building codes. The system will contain concrete pads to support condensers located outside, power disconnects, drain lines, and refrigerant lines. The home will be zoned by individual rooms and areas on each floor based on equipment specifications and load calculations. Heads for AC will be located in all bedrooms (4) and main living rooms (3) distributed optimally throughout First Floor and Second Floor areas.

23 83 00 - Radiant Heating Units

Complete installation shall conform to appropriate local codes and shall also be in accordance with manufacturer's specification. Aluminum PEX heating tubes shall not leave heated area.

Directly embed heat PEX in 3-4"-inch concrete slab located in basement floor (excluding storage closet) and garage floor for the future. Basement and garage floor heating will not be activated but will be available should these areas be insulated and finished in the future. Heat tubing will be placed directly in radiant panels throughout 1st and 2nd floors for active heating. Manifolds will be installed in basement for first floor heat and 1st floor for second floor heating at predetermined accessible mechanical closet/cubby locations for efficient distribution of hot water. DO NOT PENETRATE FLOORING FOLLOWING INSTALLATION OF HOT WATER LINES.

Division 26 00 00. Electrical

26 05 00 - Common Work Results for Electrical

From electrical pole, install service pole to receive wiring to home. Raceways to be PVC #2 Plastic Electrical conduit and/or schedule 80 steel. Where permitted by code, non-metallic sheathed cable may be used. Type THW or THWN 600 volt insulation conductors shall be used, minimum wire size shall be #12. Aluminum wire shall not be permitted. Wiring shall connect into metal recessed electrical panel, as shown on construction documents. Electrical service shall be rated at 200 amp or 400 amp residential service as determined by expected demand load calculations. Wiring from the outside meter box shall be aluminum.

26 05 19 - Conductors and Cables

Provide and install necessary circuits and breakers for appliances as stated in manufacturer's recommendations per applicable building code requirements. For general illumination, provide a minimum 15 amp circuit for each 500 sf of living area (load 3 watts per sq.ft.) Branch circuits shall be wired with No. 14 gauge wire. Install GFI circuits with No. 12 gauge wire in all wet areas, baths and exterior outlets.

All appliance circuits shall be installed per applicable building code requirements.

26 24 00 - Switchboards and Panelboards

Panelboards shall be Square "D", G.E., or approved comparable panelboard. Provide typewritten directory of circuits mounted in box. Use dead front panelboards with one-piece cabinets constructed from code gauge steel, finished with rust inhibiting primer and baked enamel finish and manufacturer's standard color.

26 27 26 - Wiring Devices

Install [beige, white, brass](#) receptacles, switches and cover plates as per code requirements. A walkthrough with Buyers, Builder, and electrician will be scheduled prior to rough-in commencement to ensure convenience outlets and light switches are installed according to Buyer's preferences. Additional devices including switches and receptacles may be added for convenience on a per unit cost basis. For exterior receptacles install gray shingle box cover plates. When two or more switches or receptacles are located together, gang with one common. All receptacles shall be placed according to code requirements.

26 50 00 - Lighting

Provide necessary circuits and wiring for light fixtures as listed below. All lighting shall be switched as per code requirements for safety first and foremost. As similar to Wiring Devices personal preferences may be incorporated into the build as identified during walk-through with Buyers, Builder, and Electrician following framing for lighting preferences and placement of lighting fixtures including recessed lights. Fixture lists are listed below.

Special Items - Provide necessary receptacle requirements and wiring for additional items as listed below. Locate as per code and convenience locations as per scheduled walkthrough with Buyers.

Undercounter lighting: install LED lights under all upper cabinets in Kitchen and Pantry areas.

Bath Vent Fans - install fans in all bathrooms. Panasonic programable Whisper Green for whole house ventilation will be installed in at least two bathrooms.

Thermostat/Control - install for all mini-split wall units at each location and radiant digital control on each floor with exterior reset.

Smoke, Heat, & CO Detectors - install per fire code at bedrooms and common areas.

Doorbell - located at front door with chime centrally located.

CATV - located in Great Room, Kitchen, Master Bedroom, and bedrooms. Precise locations for low voltage receptacles during walkthrough. [Wireless data signal distribution and devices utilized to boost signal available upon request at additional cost along with speaker wire, and other low voltage wiring for home entertainment and communication purposes.](#) Telephone lines tied in by phone company or private contractor of your choosing following move in.

Garage Door/ Overhead Door opener - locate ceiling connection for door lift device and push button at entries by LiftMaster.

Security System - [pre-wire & security system will be installed per specifications of provider and monitoring service of choice from Buyer including Vivint, ADT, or Simplisafe. Both hardwired and wireless options available.](#)

Fixtures - Provide and install electrical fixtures in general location listed below as selected by Buyer with an allowance of \$7,500.00. Additional allowance has been made for 40 recessed lights to be installed in combined 1st and 2nd floors. Precise locations to be determined during design meeting and at walk-through with Buyer following framing and partitioning. Recessed light fixtures by wafer. Exact location of fixtures noted below also to be determined at walk-through following framing.

INTERIOR:

Mudroom
Kitchen Island
Great Room
Dining Room
Sun Room
Master Bedroom
Master Bath
Stair Hall
Powder Room
Top of Stairs to Basement
Wine Cellar
Top of Stairs to Second Floor
Bath 1
Bath 2

EXTERIOR

Garage Doors (2)
Front Entry
Mudroom Entry
Covered Terrace

Division 28 00 00. Electronic Safety and Security

28 31 46 - Smoke Detection Sensors

Install at least one smoke alarm on every floor of a structure (including the basement) and outside each sleeping area per MA fire code. Smoke alarms are required in all sleeping rooms, according to the NFPA 72, *National Fire Alarm Code®*. Smoke alarms mounted on ceilings and should be installed at least twelve inches away from the nearest wall. Mount alarm at the highest point of the ceiling. No smoke alarms installed near windows, outside doors, or ducts where drafts might interfere with their operation. Do not paint, apply finish or obstruct smoke alarms.

Division 31 00 00. Earthwork

31 10 00 - Site Clearing

The area of clearing shall be maintained within the limits shown on the appropriate plot plan. Stumps will be removed and matted roots to a depth of 18 inches below existing ground surface. Dispose of trees and shrubs in accordance with applicable garbage, refuse or weeds ordinance. No materials burned on site.

Soil Bearing - Foundation designs are based on a soil bearing value of 2500 psf. Foundations and slabs are designed to uniformly bear on well-compacted, well-drained non-expansive soils. A certified soils engineer shall review foundation designs and building loads and compare with subsurface soil investigation as needed. .

31 11 00 - Clearing and Grubbing

Clear and grub the construction site off Ware Lane with additional care given to mature oak tree protection. Remove dead or dying trees along Ware Lane as marked with yellow tape. Grade building site with appropriate soils. Existing trees to remain unless otherwise requested to be removed by Buyer or are in the footprint or too close to the new house. Trees shall be marked prior to clearing and/or protected to prevent damage. If any damage is done to walkways, driveways, etc, needed repairs shall be provided by the Builder. Repair or replace any damaged vegetation or terrain that is indicated to be

protected or is more than eight feet from the edge of any construction.

31 20 00 - Earth Moving

Excavate bottom of all foundation walls and footings at building perimeter a minimum of 48" below frost line and 24" wide. Base of footings shall extend down to undisturbed virgin soil which has been compacted to 95 percent proctor density. All excavation shall be to a level below existing demolition debris. Board form all footing as required by soil conditions.

For basement walls, excavate area indicated on construction documents, allowing an additional 18-24" minimum clearance around the perimeter of foundation walls for proper drainage and waterproofing assembly.

At slab foundations, compact sub-grade under slabs to a minimum 95% density. Compact backfill areas not under slabs or foundation to a minimum 90% ASTM D-689. Sub-base directly under concrete slabs on grade shall be a minimum of four inches of compacted granular fill.

31 22 00 - Grading

Carefully remove loam and topsoil to be incorporated in the finished work and store separate from the other excavated material. Failure to isolate loam and topsoil from the other excavations shall require that said soils not be used as topsoil.

When excavations are to be made in paved surfaces, remove pavement so as to provide a clean, uniform edge with a minimum disturbance of remaining pavement. Dispose large pieces of pavement away from the site of the work immediately.

31 22 13 - Rough Grading

Prior to commencement of earthwork, perform such soil and rock removal and filling as may be required to facilitate the progress of the work and bring all elevations to the rough grade lines indicated on the Contract Documents. Fill or backfill as required.

31 22 19 - Finish Grading

Keep exterior finished grade a minimum of 8" below finished floor elevation by backfilling with appropriate soils. Provide swales with positive outfall and slope grade away from building to allow water to drain away from the building foundation. Do not backfill against foundation until project is completely framed and roof structure is in place.

31 23 00 - Excavation and Fill

Backfill material to be used from the excavations shall be of such nature that after it has been placed and properly compacted, it will make a dense, stable fill. It shall not contain vegetation, masses of roots, stones over 3-inches in diameter, or porous matter and shall not be saturated. Organic matter shall not exceed minor quantities and shall be well distributed.

31 23 16 - Excavation

Carry out the excavation, dewatering, sheeting and bracing in such manner as to eliminate any possibility of undermining or disturbing the foundations or any existing structure or any work previously completed.

Excavate pipe trenches to the necessary depth per code. Trenches over 5 feet in depth shall be properly sloped, shored, braced or otherwise supported in conformance with the OSHA Construction Safety and Health Standards.

Excavate trenches to provide a uniform and continuous bearing and support for the pipe and appurtenant structures on solid and undisturbed ground and at the specified grade at every point.

Excavation for structures and pipelines shall include the disposal of materials unsuitable

for reuse for backfill. Excavation activities shall include the stockpiling of suitable materials which shall be incorporated into the project at a later date of different location.

31 23 23.13 - Backfill

Correct any part of the trench bottom excavated below the specified grade with approved materials and thoroughly compact.

Complete all backfilling to the dimensions and levels shown on the construction documents. Where excavated material or any portion thereof is deemed unsuitable for backfilling material, procure and place approved select borrow materials.

Backfill as promptly as is consistent with non-damage to the installed structures. Do not place frozen material in the backfill.

No material shall be placed or compacted when it is too wet or frozen or when the sub-grade or previously placed material is too wet or frozen.

31 25 00 - Erosion and Sedimentation Controls

Clear the top layer of soil and place in a designated area for use at the end of the project. Provide swales with positive outfall, and slope grade away from building to allow water to drain away from the foundation. Backfill around building with subsoil graded free of lumps larger than 6", rocks larger than 3" and debris. Keep finished grade elevations a minimum of 8" below finished floor elevation. Do not backfill against foundation, until home 1st floor deck is framed.

31 31 16 - Termite Control

If required, Foundations shall be pre-treated under all slabs and crawlspace areas between vapor barrier and granular sub-base to conform with HUD minimum standards and applicable building codes. Treatments shall not be made when soil is excessively wet or after heavy rains. Contractor shall provide a one-year renewable warranty.

31 40 00 - Shoring and Underpinning

Existing footings, foundations, pile caps, grade beams, retaining walls or pavement which may be affected by excavation operations shall be shored or underpinned adequately or otherwise protected against settlement and shall be protected against lateral movement. Provide necessary materials to hold back earth at excavations and as required to prevent cave-ins and earth sloughs.

31 50 00 - Excavation Support and Protection

Install excavation support systems for safety preservation of existing improvements. Design criteria of support systems shall consider all loads in a manner which will allow the safe and expeditious construction of permanent structures without movement or settlement of the ground.

Division 32 00 00. Exterior Improvements

32 01 90 - Operation and Maintenance of Planting

All plants shall be planted and kept in healthy, growing condition by watering, weeding, cultivating, pruning, spraying, trimming, protection from wind, and by performing any other necessary operations or maintenance for a period of 30 days prior to closing or until acceptance of the planting at the time of the final inspection. A final weeding of all plant areas shall be made immediately prior to final inspection. Newly planted trees shall be cared for and pruned as necessary. All dead branches shall be removed. Rootstock shoots from grafted material shall be removed.

32 01 90.13 - Fertilizing

Soil conditioners and natural additives including Coast of Maine, cow manure, etc will be incorporated in planting areas and sod applied to heavily disturbed areas by clearing operations. Soil conditioners and fertilizers will be spread as needed and thoroughly incorporate by rototilling work into topsoil to a depth of 4". Topsoil will be raked until the surface is finely pulverized and smooth with aggregate of 1/4" or smaller remaining only.

Plantings	Name - Species	Qty	Location
<i>Canopy Tree</i>	Red Maple 6'-8'	1	<i>Front yard / Driveway area</i>
<i>Understory Tree</i>	Bradford Pear	2	<i>(1) Front, (2) Back</i>
Winter gem	boxwood	24	(4) Front, (4) Back
Evergreens	Colorado Spruce & Leland Cyprus	8	South side. privacy
Canopy Tree	River Birch 8-10'	2	North east and south east corners
Understory Tree	Eastern Redbud	1	west side
Canopy Tree	Copper Beach 6'-8'	2	East side
Canopy Tree	American Beach 8'-10'	2	Ware Lane
Evergreen	Green Giant 10'-12'	12	East side. privacy
Misc	Assorted	24	Roses, salt grass, front and rear

33 06 00 – Fence

NA

32 12 16 - Asphalt Driveway

Sub-base materials shall Lynn pack stone below finished driveway. The base course shall be bituminous concrete consisting of coarse and fine aggregate combined with asphalt cement, resulting in a mixture of Type BM-2. The surface course shall be bituminous concrete consisting of crushed stone, crushed slag or crushed gravel and the fine aggregate slag or stone screenings or combination thereof combined with asphalt cement, resulting in a mixture of Type SM-2A.

Asphalt Installation - Areas to be paved shall be covered with a layer of hot asphalt concrete surfacing not less than the thickness indicated after compaction (ex: compacted thickness shall be 2 inches binder and 2 inches of finish, driveways and aisles of parking areas). Paving asphaltic concrete shall be delivered, laid, rolled and finished in accordance with local guidelines. Finish paving shall conform to slopes, lines and finish grades as, and drain properly.

*Paver driveway upgrade option available at additional cost. See Builder for options if interested.

32 13 13 - Concrete Pads and Walks

Provide light duty paving at exterior walk areas consisting of a 2" thick pavers or slab on sub-grade compacted to 98 percent density with 3000 psi concrete. A 2" thick thermal bluestone landing will be installed at front entry and side porch area. Back porch/terrace will be thermal bluestone installation. Refer to exterior landscape plan for additional information on walk areas for design. Expansion joints shall be installed as in standard concrete practices for all concrete walkways. Control joints shall be installed at pre-determined locations no later than 12 hours after installation.

32 14 00 - Paver Terrace & Front Walk. Bluestone Steps.

Install thermal bluestone at entry, side entries, and rear terrace area over concrete. Pitch shall be such that final installation will result in proper surface drainage. Set with mortar joints. After installation wash access mortar from joints.

Color: PA Bluestone

Finish: level with slight pitch away from house.

Bluestone/Granite terrace at landings including front entry, side entries, and rear terrace. New England fieldstone veneer risers and veneers at Entry.

32 84 13 & 23 - Drip Irrigation & Underground Sprinklers

Low pressure water system with buried or sub-surfaced pipes. Install as recommended by manufacturer.

Install irrigation system consisting of valves, water lines, sprinkler heads and control panels to adequately cover sod and plant beds. Install as recommended by manufacturer. electronic valves. Irrigation water deduct meter will be installed in the basement besides regular water meter.

Trenching and Backfill - All main lines shall be buried a minimum of 6" inches, and shall be covered with sand to six (6") inches above main. Laterals shall be buried a minimum of twelve (8") inches for spray heads and a minimum of eighteen (18") inches for rotors.

Piping Installation - Piping shall be laid parallel in trenches and shall not cross over or wrap around each other and shall be buried at least one (1") inch apart. Cutting and gluing of PVC pipe shall be done only with appropriate equipment and methods recommended by manufacturers of PVC solvent cement and PVC pipe.

Connections - Connections to existing water mains, other than at valve outlets, shall be made with compression tees and gate valves. All pipe under concrete, asphalt, or other masonry shall be cased loosely in larger size pipe with at least six (6") inches protruding from the sleeve before fittings are attached. Earth shall be compacted under the pipe where it extends from sleeve. All fittings shall have at least a two (2") inch clearance from other pipes or fittings.

Joints - All solvent cement joints shall be made with medium-body cement or heavy-body only with a pre-coat of primer. Excess cement shall be wiped off all joints. Swing joints shall be sealed with Teflon tape on threads. All other threaded joints shall be sealed with a non-drying thread joint compound.

Deduct Meter Installation & Backflow prevention - Before installation of remote control valves, quick coupling valves, or sprinkler heads, all lines shall be flushed with water. All remote control valves shall be installed in large rectangular plastic boxes flush to grade unless there is a special need for sub-surface installation, in which case valve markers shall be used with access to the flow control handle.

Valves - All quick-coupler valves shall be installed in round plastic boxes flush to grade. Anti-drain valves shall be installed on all low sprinkler heads. All electric remote control valves shall be installed with a gate valve on the pressurized side and a PVC union on the non-pressurized side. [All remote control valve boxes shall be marked with a code system and sequenced in the controller in a logical manner to facilitate maintenance, i.e. A-1, A-2 \(clock A, valve 1 and 2\).](#)

Sprinkler Head Installation - All sprinkler heads shall be attached to double swing joints. Shrub heads on risers shall be staked with 1/2 inch rebar as needed.

Sprinkler Heads - All sprinkler heads two (2') feet or less from lateral connection shall be connected by threaded fittings only to the lateral. All sprinkler heads shall be installed with head flange at least 1 inch from header. In the case of shrub heads on risers, the riser shall have a minimum [two \(2"\) inch](#) clearance to the header. Sprinkler heads of differing precipitation rates shall not be valved together. Shrub heads located along pathways or traffic areas shall be the type that retract to ground level when watering is completed; Toro, or approved equal. All shrub, ground cover and lawn heads shall be spaced so as to spray from head to head as minimum, preferred coverage is 1.25 to 1.5 times spacing.

32 91 13 - Soil Preparation

Planting will not commence until finish grades are established and planting areas are properly prepared and graded. When preliminary grading, including weeding and fertilizing, has been completed and the soil may be readily worked, grade all planting areas to a smooth, even and uniform plane with no abrupt change in surface. Slope soil areas adjacent to buildings away from the buildings, and direct surface drainage as indicated on the drawings. Grading shall provide for natural runoff of water without low spot or pockets. Finish grade of earth in landscaped areas shall be [3 1/2 inches](#) below the top of adjacent pavement, curbs or headers.

32 91 13.16 - Mulching

Apply black mulch immediately prior to seeding in all planting beds. Loosen baled straw and thoroughly break up before placing over seeded area. Begin placement of mulch on the windward side and from the toe to slopes. On slopes 2 to 1 and greater provide jute matting stapled [18 inches to 3 feet](#) apart using closer spacing around curves and areas of concentrated storm water runoff. Soil amendment and mulch shall consist of Cedar-soil or equal, composted, nitrogen-stabilized, water-holding materials with long residual life.

32 91 19.13 - Topsoil Placement and Grading

A **three (3") inch** cover of topsoil or appropriate soil amendment shall be spread uniformly over the soil and tilled into the top four (4") inches of soil. Topsoil shall be a natural, fertile, friable soil, typical of productive soil in the vicinity, obtained from naturally well drained areas. Rototill all areas indicated on plans and on areas damaged by construction, to depth of 4", removing stumps, all foreign objects and stones larger than one inch diameter. Place topsoil on all areas and incorporate by rototilling into subsoil.

32 92 19 - Seeding

Seed only when weather conditions are suitable at side and rear yard areas. All newly seeded turf areas shall be free of broadleaf weeds. Sow seed with mechanical spreaders at the specified rate on a calm day. Sow one half the seed in one direction and the other half at right angles. Seed shall be raked lightly into the soil to a depth of 1/4 inch. Keep the surface moist by a fine spray until the grass shows uniform germination over the entire area. Wherever poor germination occurs in areas larger than three (3) square feet, reseed, roll and water as necessary to obtain proper germination. Infested areas shall be treated with a selective broadleaf insecticide; Trimec or approved equal.

32 92 23 - Sodding

Plant only certified sod in front yard areas facing Ware Lane when the soil is moist and favorable for growth. Shape the area to be sodded and finish to the lines and grades indicated on the plans. Loosen the surface prior to placing sod. Keep the grade moist by sprinkling, if necessary, sod on the prepared surface with the edges in close contact. Each piece of sod laid shall be fitted and tamped into place with hand tampers. Apply a sufficient quantity of water to all sod after laying and to prevent the sod from drying out for a period at least two weeks to ensure growth.

32 93 00 - Plants

Shrubs and trees shall be of a variety, size and quantity per above planting list. Cut burlap, twin and wire baskets from top 12 inches of root-ball and remove from site. Backfill with 1/2 clean existing soil, 1/4 sand and 1/4 peat moss and manure. Plants shall bear some relation to soil level when planted as they did when in container. Place each plant in center of plant pit. Firmly tap backfill material into plant pits around and under the root ball to force out all air pockets. Backfill as specified on the plans. Thoroughly water to saturate the root ball and backfill. Stake all trees with hardwood stakes driven into firm ground and secure tree to stake.

32 93 23 - Plants

Hand dig holes without undermining existing facilities. Do not loosen soil at the bottom of hole, compact soil under root ball if loose. Dimensions of the hole will vary with the size of the root ball. Tree wells located adjacent sidewalks, shall have the top of root ball four inches (4") below the sidewalk surface. Place each plant in an upright and plumb position.

Berms - A berm of earth approximately four (4") inches high shall be built around the perimeter of the backfilled plant holes to form a shallow basin for the retention of irrigation water. Each basin shall be mulched with two (2") inches of mulch.

Backfill - Backfill shall consist of the excavated material with all stones larger than 1/2" removed, and with soil amendment added in the proportion of one (1) part amendment to three (3) parts soil thoroughly mixed. If additional soil is required, site topsoil shall be used. The backfill mixture shall be finely divided, loose, and free of clods.

Division 33 00 00. Utilities

33 00 00 - Utilities

Install necessary utility services, such as electricity underground, water, gas, sanitary sewerage and support structures for power and communications. Coordinate requirements with local utility providers. All utilities shall be located underground from street to building, unless otherwise stated.

Seller	Date
--------	------

Buyer	Date
-------	------